Technische Universität Dresden

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Data Visualization

2 – Perception and Cognition

Computergraphik und Visualisierung

INTERACTIVE MEDIA LAB DRESDEN

Faculty of Computer Science – Institute of Software and Multimedia Technology

Perception & Cognition
Lecture Outline

1. Visual Perception – Introduction
2. The Visual System
3. Cognitive Processing & Pattern Recognition
4. Preattentive Processing
5. Gestalt Principles
6. Perception of Color and Contrast
7. Summary and Outlook

Perception and Cognition

(1) Visual Perception – Introduction

TUD

Visual Perception
Human Vision

- Highest bandwidth sense
- Fast, parallel, processes large amounts of data
- Permits perception of
  - Colors, textures, movements, objects
  - Relations between objects and groups of objects
- Pattern recognition
- Pre-attentive processing
- Extends memory and cognitive capacity
- People think visually
- Impressive. Lets use it!



The eye. . .
the window of the soul,
is the principal means
by which the central sense
can most completely and
abundantly appreciate
the infinite works of nature.

Leonardo da Vinci (1452 – 1519)

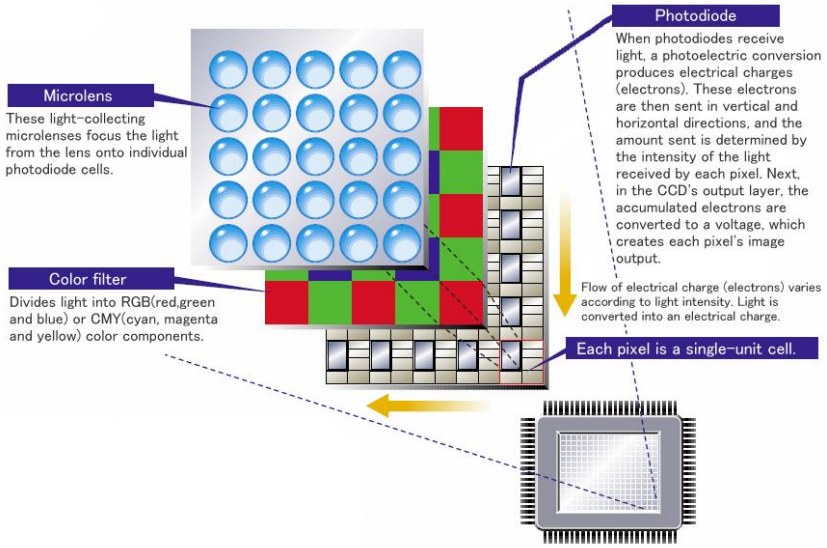
Effective data presentation is rooted in an understanding of visual perception.

[North09]

Visual Perception
Our Visual System – Eyes vs. Cameras

Cameras

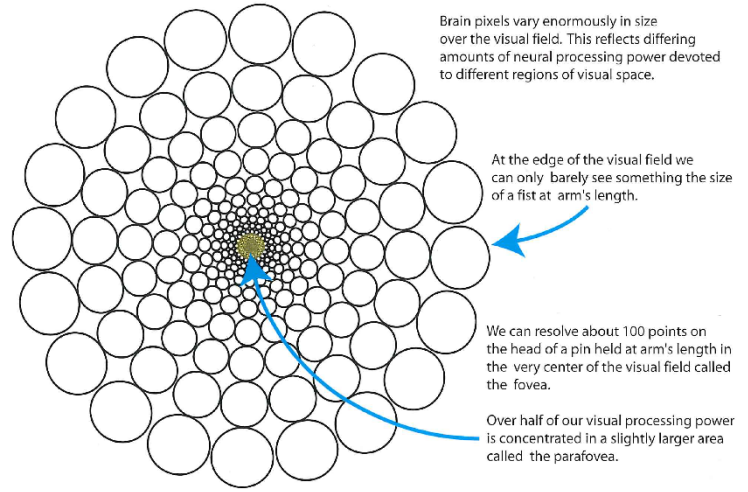
- Good optics
- Single focus, white balance, exposure
- “Full image capture”



Source: http://av.jpn.support.panasonic.com/support/global/cs/dsc/knowhow/knowhow27.html

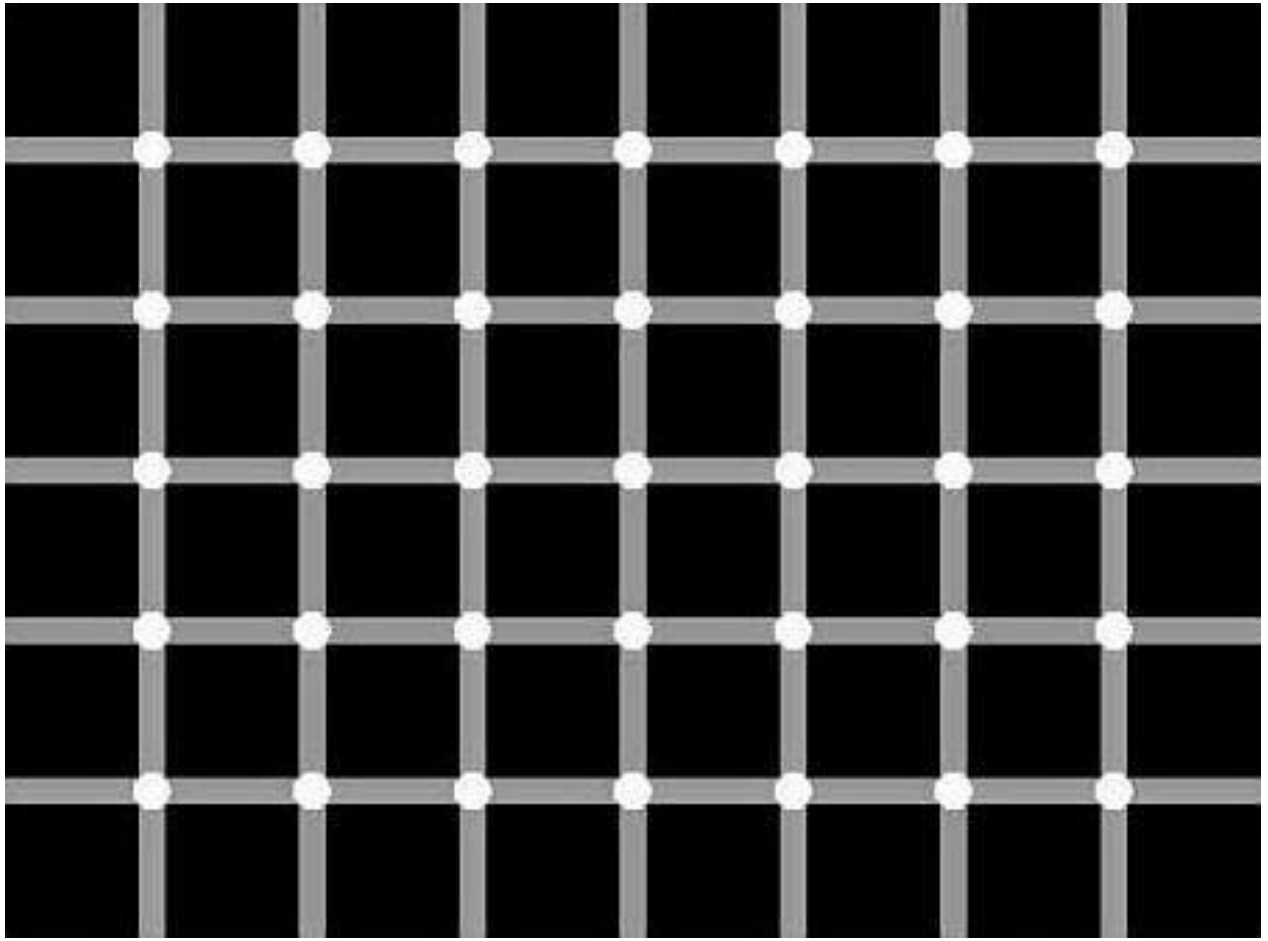
Eyes

- Relatively poor optics
- Constantly scanning (saccades)
- Constantly adjusting focus
- Constantly adapting (white balance, exposure)
- Mental reconstruction of image (sort of)

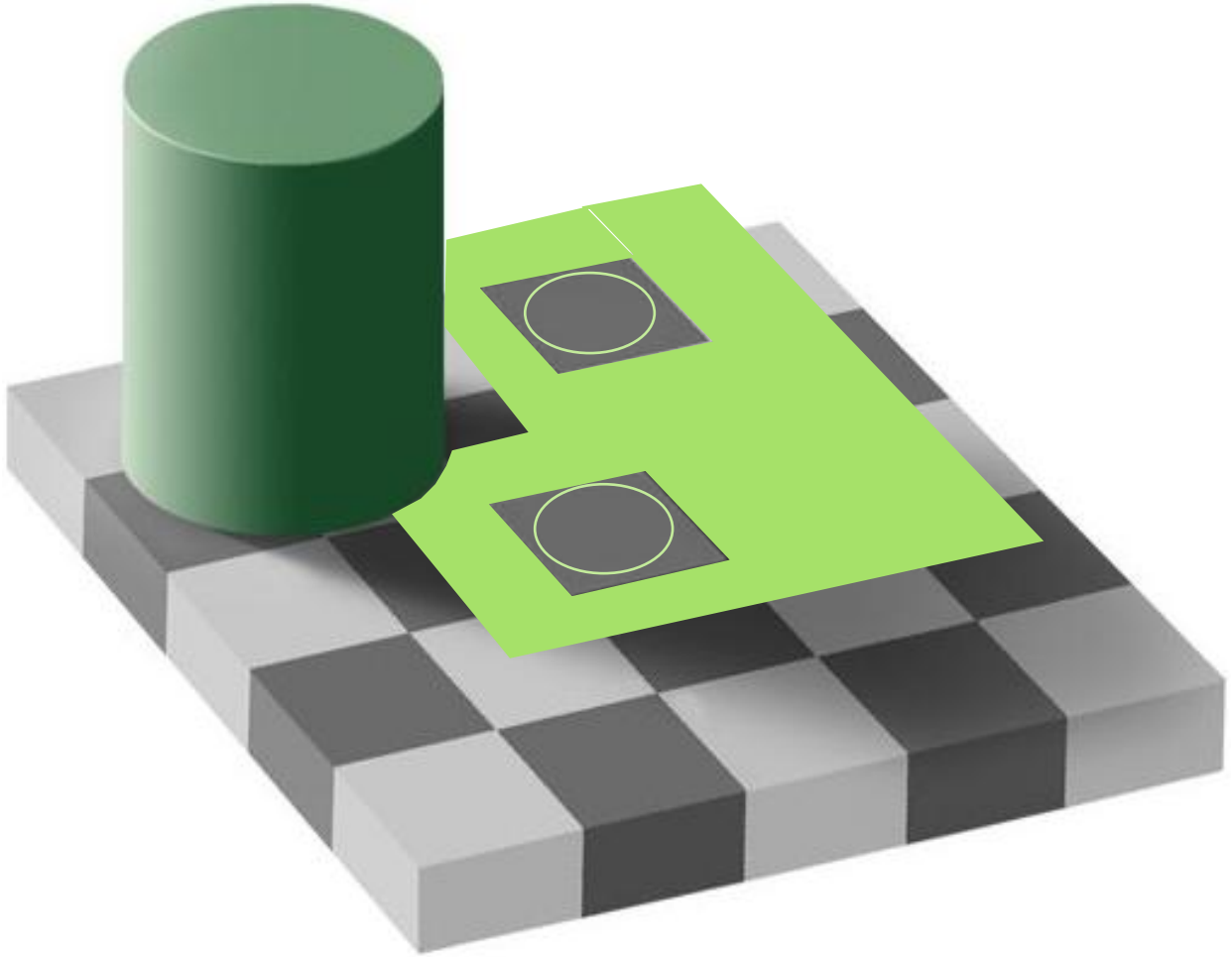


[Ware08]

Visual Perception
… is more than camera work

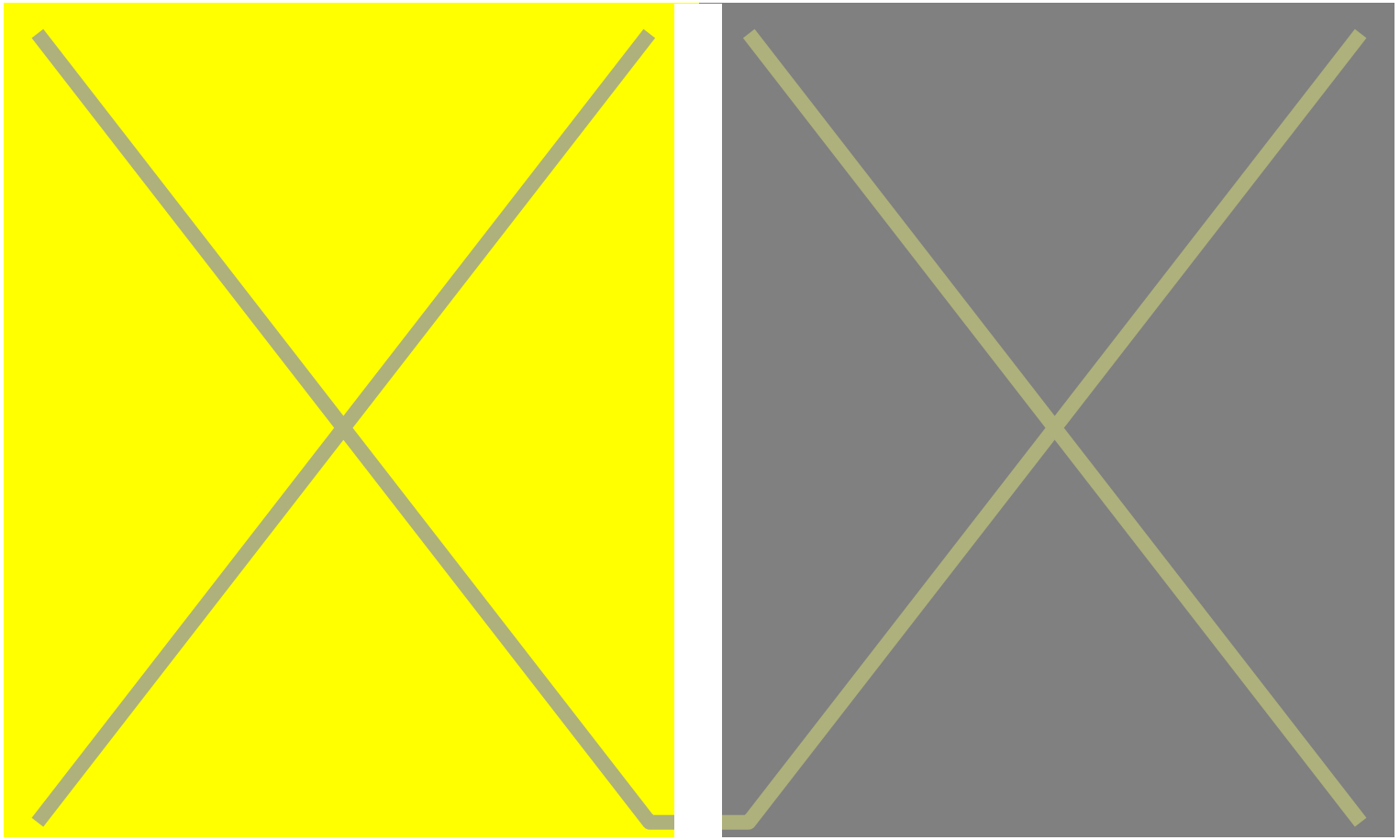


Visual Perception
… is more than camera work



Visual Perception
… is more than camera work

Even color is relative



Visual Perception
… is more than camera work

More examples: Akiyoshi Kitaoka’s illusions



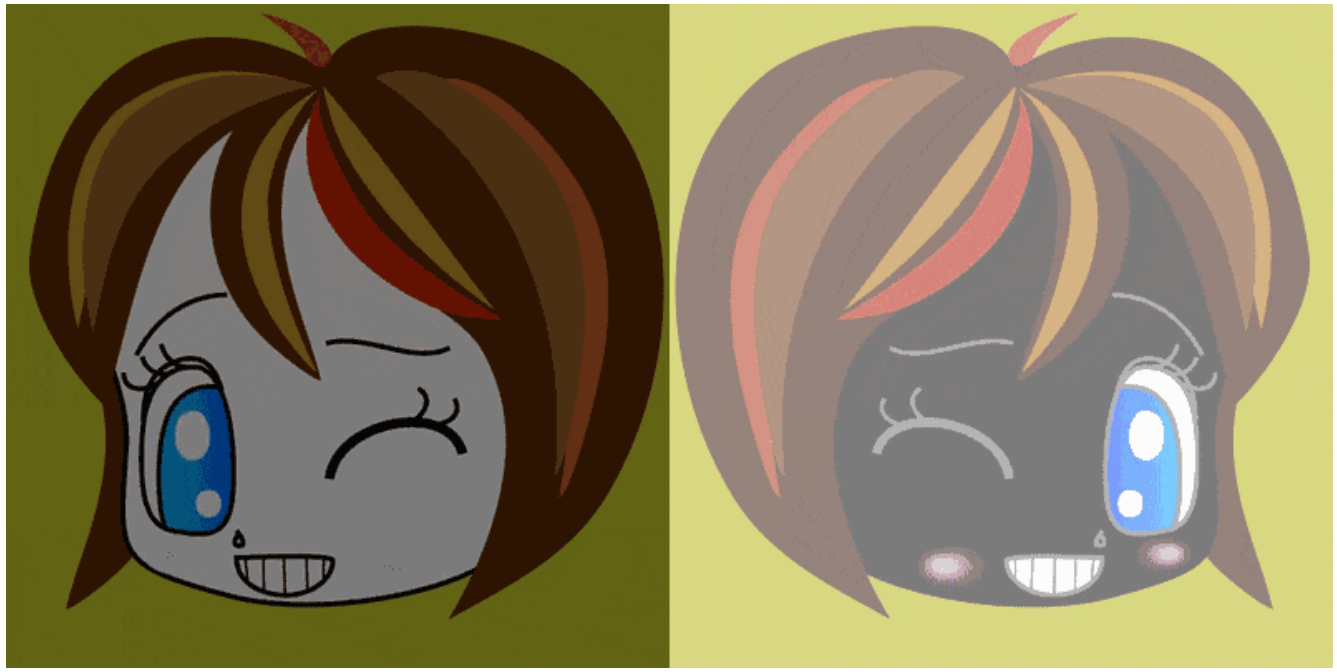


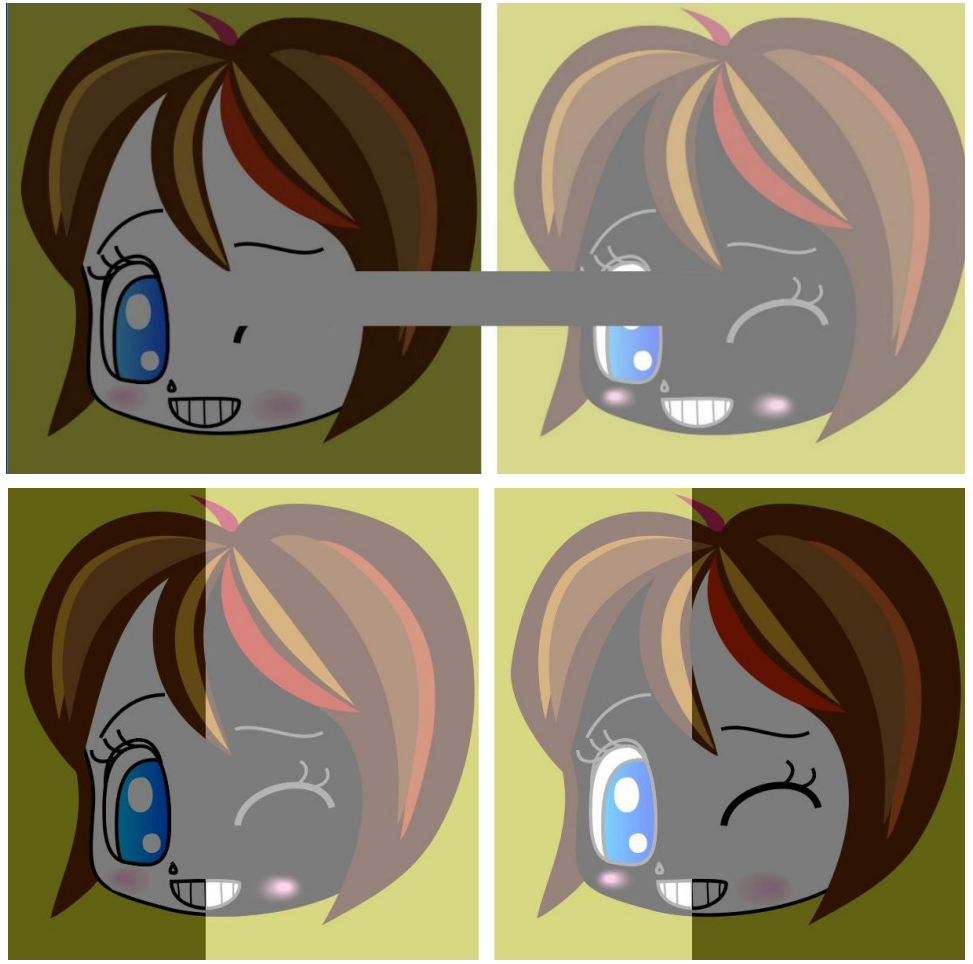
Original thread:
https://x.com/AkiyoshiKitaoka/status/1798705648001327209

Check out more of Akiyoshi‘s work:
https://www.ritsumei.ac.jp/~akitaoka/index-e.html

Visual Perception
… is more than camera work

Both faces have the same luminance!





Source of .gif and article explaining the effect:
https://wolfr.am/1n4af9omx

Check out more of Akiyoshi's work:
https://www.ritsumei.ac.jp/~akitaoka/index-e.html

Visual Perception
… is more than camera work

We even perceive the written word differently than you probably think

Aoccdrnig to rscheearch at Cmabridge Uinervtisy, it deosn't mttaer In what oredr the ltteers in a word are, the olny iprmoatnt tihng is that the frist and lsat ltteer be at the rghit pclae. The rset can be a Total mses and you can still raed it wouthit problem. This is bcuseae the huamn mnid deos not raed ervey lteter by istlef, but the word as a wlohe.

amzanig huh?

Visual Perception
… is more than camera work

- Quickly name the colors of the words!

RED
GREEN
PURPLE
ORANGE
BLUE

[Hearst08]

Visual Perception
Vision as Knowledge Acquisition

Perception as a Constructive Act

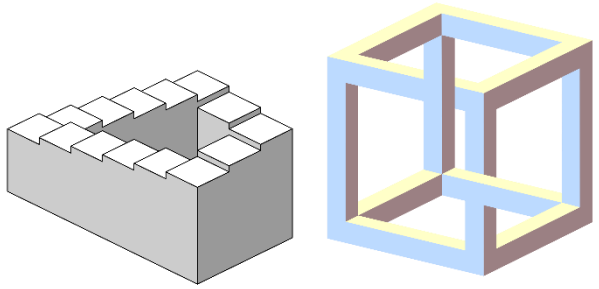
- What you see is not necessarily what you get
  - Adaptation of vision to different lighting situations
  - Image aftereffects
  - Optical illusions
  - Ambiguous figures



https://www.cleareyes.com/eye-care-blog/201610/types-optical-illusions/

Perception as Modeling the Environment

- Evolutionary purpose
- When you close your eyes, the world doesn't disappear!
- Examples: Visual completion, Object occlusion, Impossible objects



https://de.wikipedia.org/wiki/Unmögliche_Figur

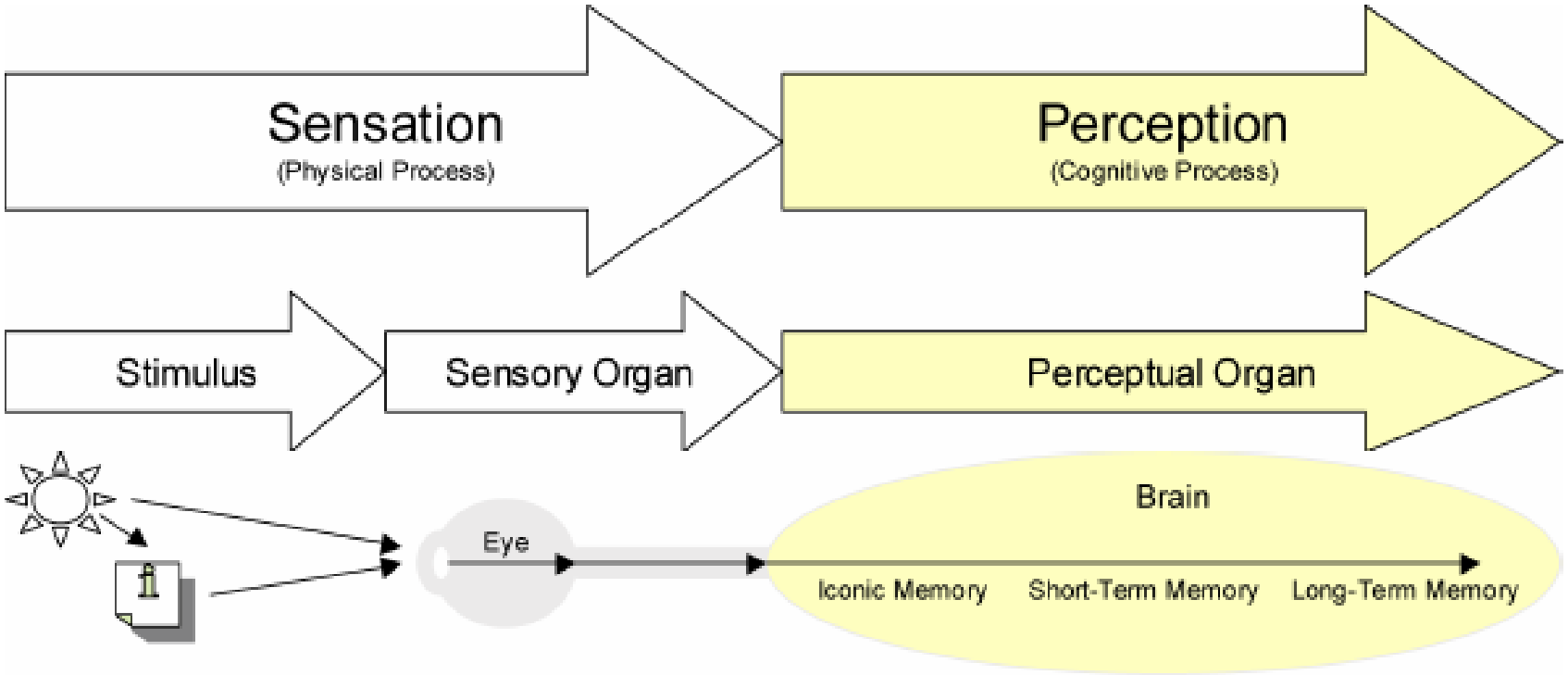
Perception as Apprehension of Meaning

- Classification
- Attention and consciousness

[Hearst08]

Visual Perception
Seeing as Constructive Action

An effective data visualization is rooted in a good understanding of visual perception.



[Stone07]

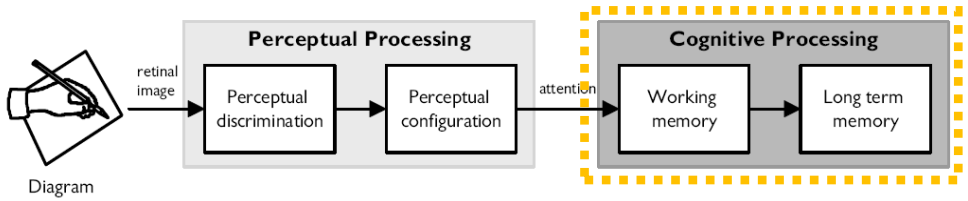
Perception and Cognition

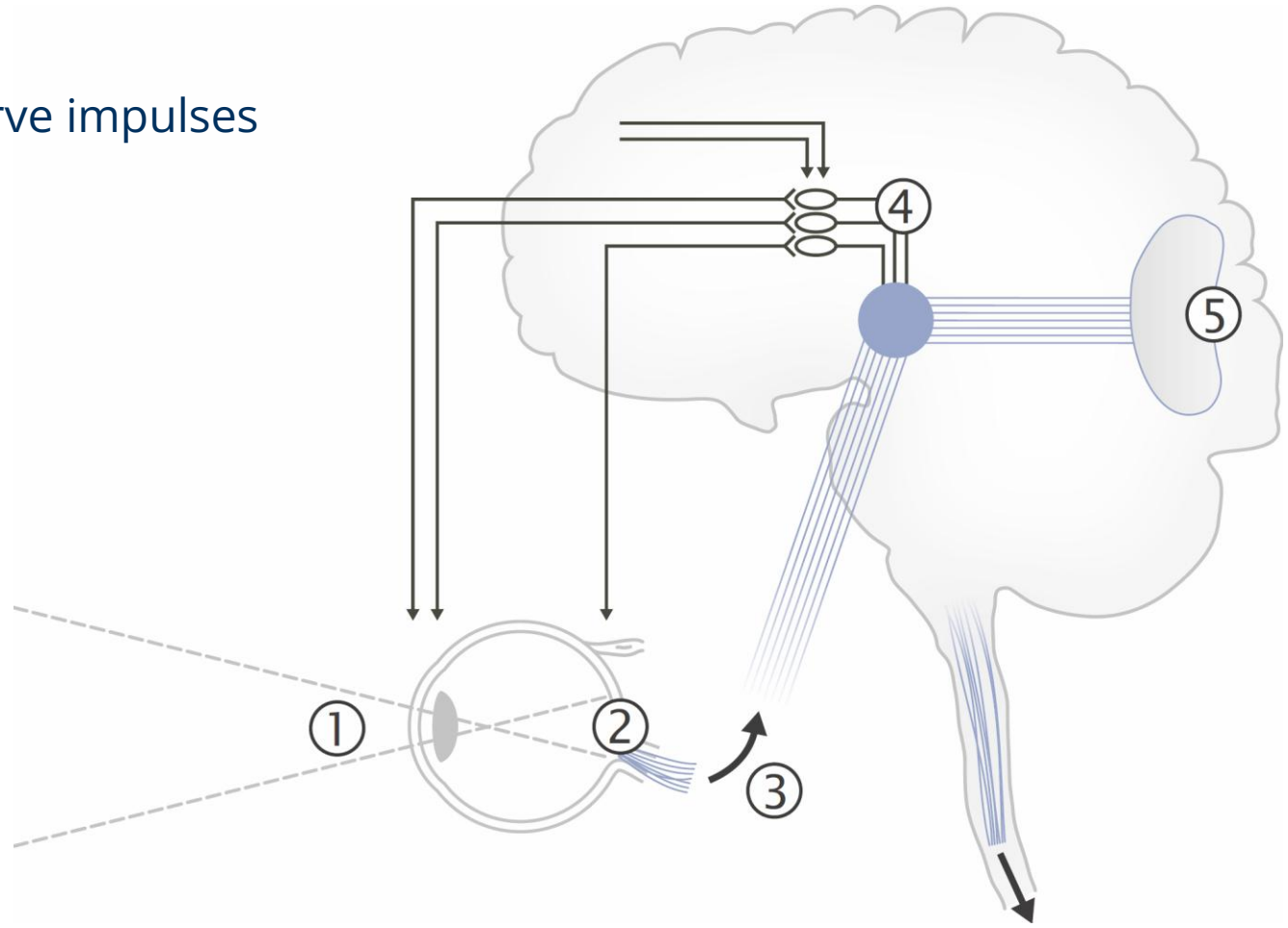
(2) The Visual System

TUD

The Visual System
Stepwise Processing of Visual Information

1. Transformation of incident light in nerve impulses
   - fast parallel processing
   - data -driven, bottom-up
   - extraction of features (contours, movements, orientation)
2. Transmission and first separation of the stimuli into nerve tracts
3. Interpretation of stimuli in the visual cortex (recognition)
   - Interpretation is top-down, involves comparisons with short and long term memory

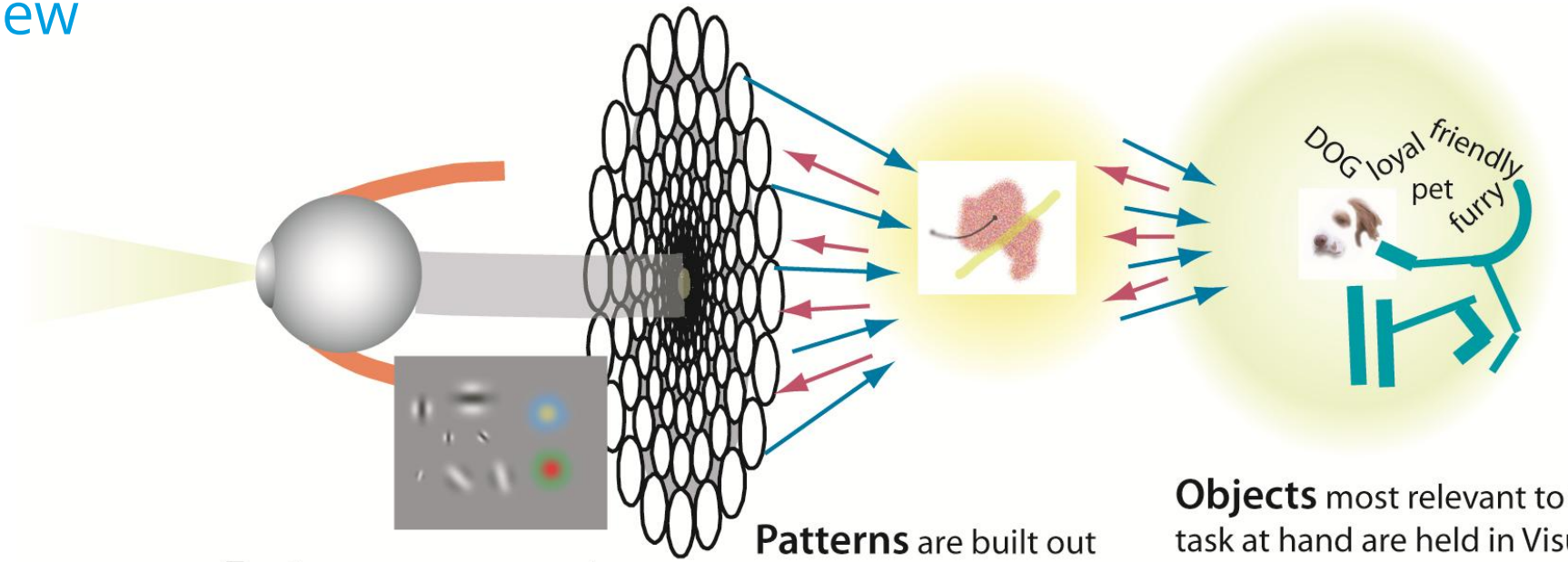




1 Lens, 2 Retina, 3 Optical nerve, 4 Kniehöcker (lateral geniculate nucleus), 5 Visual cortex

[Lehmann et al. 97]

The Visual System
Overview



Features are processed in parallel from every part of the visual field. Millions of features are processed simultaneously.

Patterns are built out of features depending on attentional demands. Attentional tuning reinforces those most relevant.

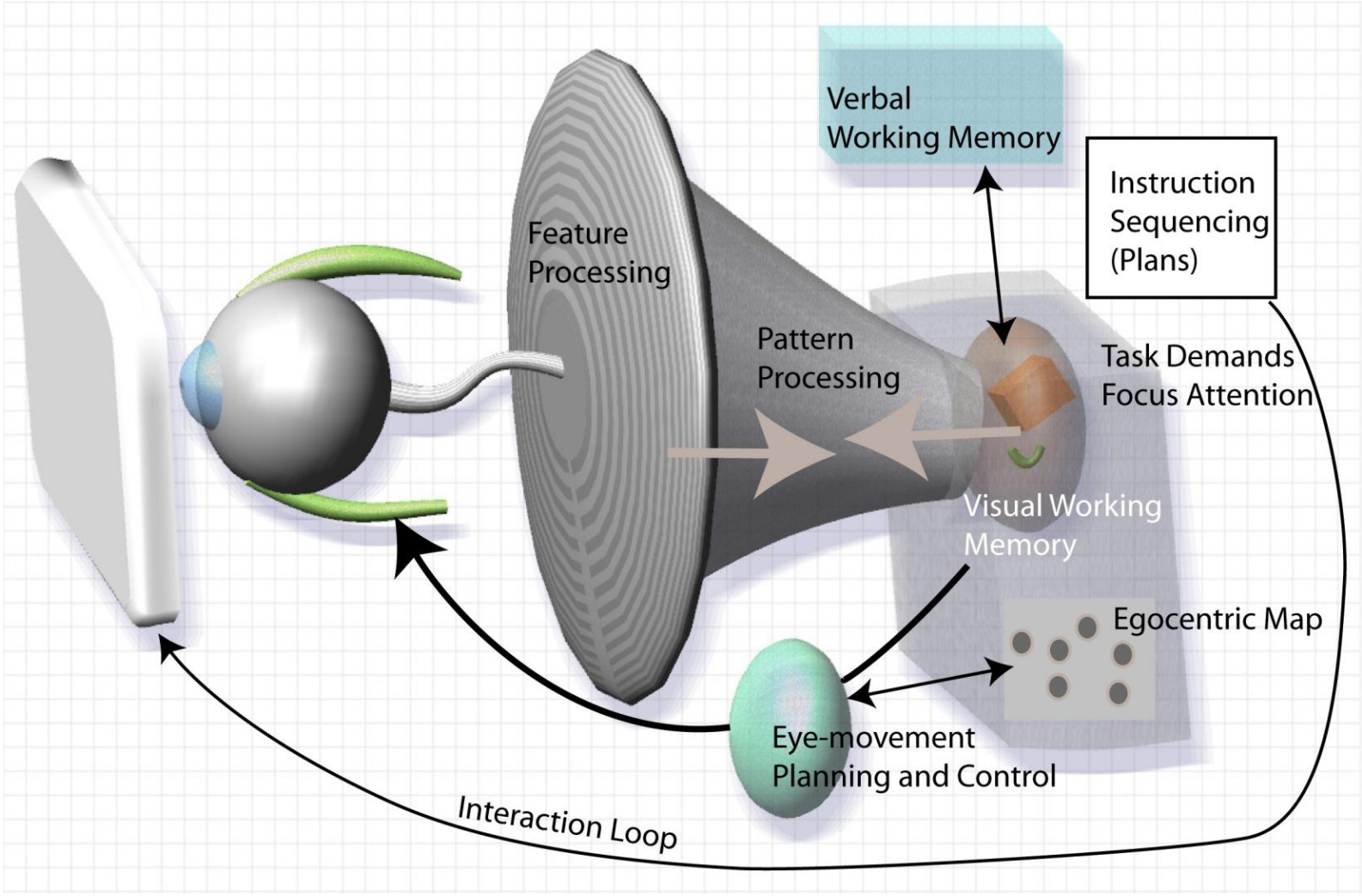
Objects most relevant to the task at hand are held in Visual Working Memory. Only between one and three are held at any instant. Objects have both non-visual and visual attributes.

Bottom-up information drives pattern building

Top-down attentional processes reinforce relevant information

[Ware08]

The Visual System
Overview



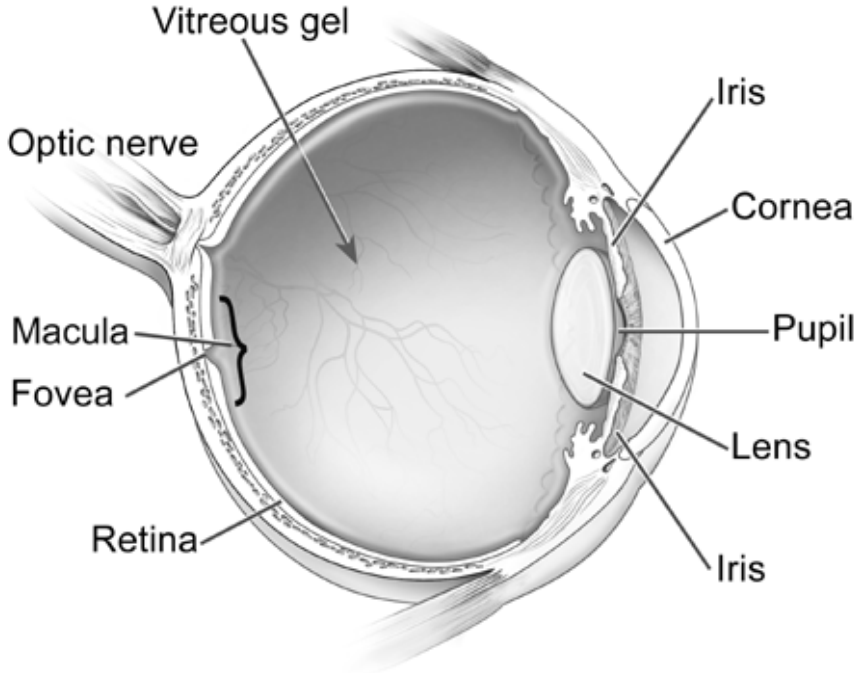
[Ware08]

The Visual System
Our Eye

Optical system of the eye consisting of the cornea (dt. Hornhaut), iris, lens and vitreous body (dt. Glaskörper).

Analogy to a camera

- Optical system corresponds to an objective
- Retina corresponds to a film



[Source: National Eye Institute, https://www.nei.nih.gov/]

The Visual System
Our Eye



Two kinds of light-sensitive cells (dt. Photorezeptoren)

- Rods (dt. Stäbchen) (≈ 120 Millions) for recognizing grey values, very light sensitive
  - response to a stimulus ≈ 300 ms
- Cones (dt. Zapfen) (≈ 6,5 Millions) for recognizing colors
  - response to a stimulus ≈ 80 ms
  - 3 kinds which are sensitive to different wave length (red, green, blue with frequency 10:10:1)
    → Consequence for color scales

Irregular distribution of light-sensitive cells

- Fovea: high resolution visual perception, almost exclusively cones
- Periphery: almost exclusively rods (color blindness in the periphery), low resolution

The Visual System
Our Eye

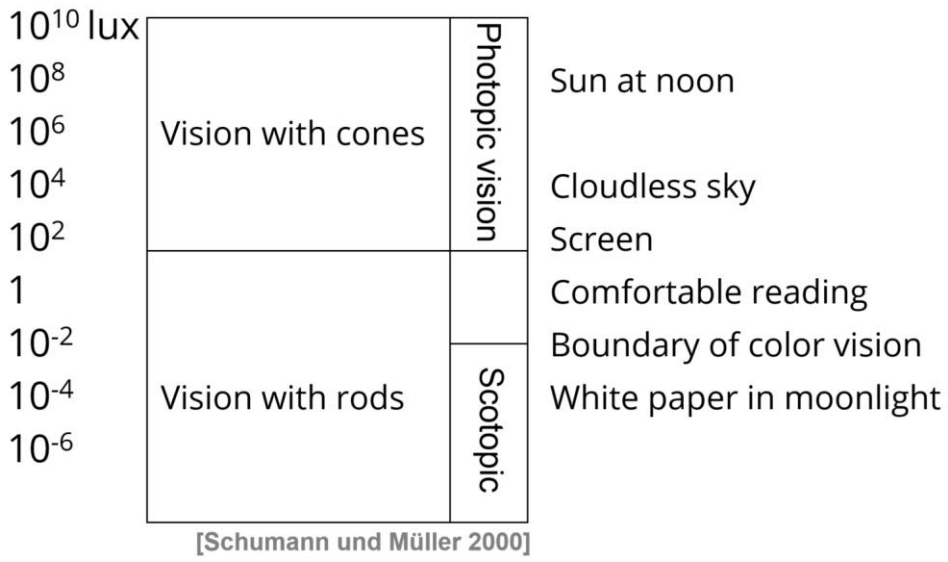


Spatial resolution of the human visual system

- Width of cones ~ 3 µm → Maximum spatial resolution: 50 arc seconds (50”).
- Called visual acuity (dt. Punktsehschärfe)
- Perception of discontinuities of line structures below the visual acuit

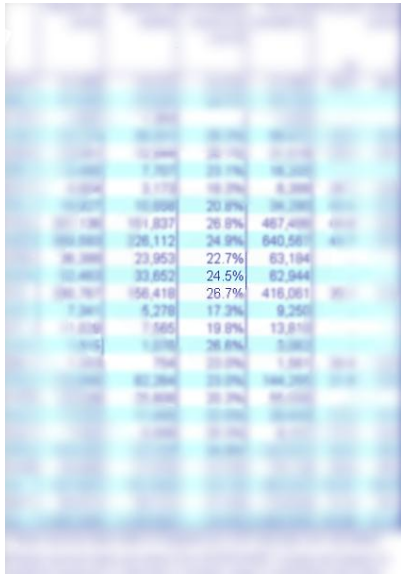
Consequences:

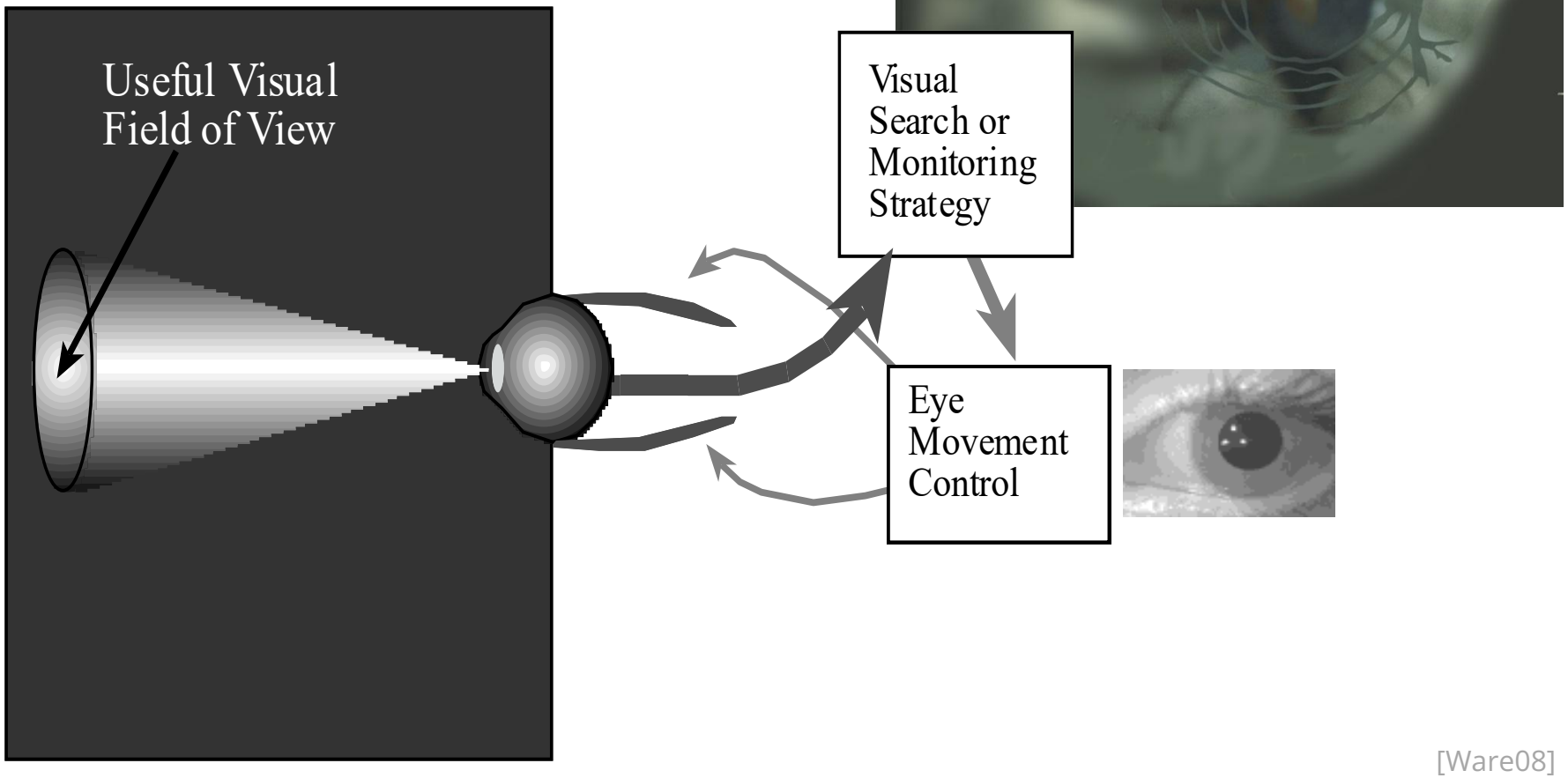
- Perception occurs in different levels of detail
- Under low illumination
  - Visual perception through more light sensitive rods
  - Almost no color perception,
    strongly reduced spatial resolution,
    recognition decelerated



[Schumann und Müller 2000]

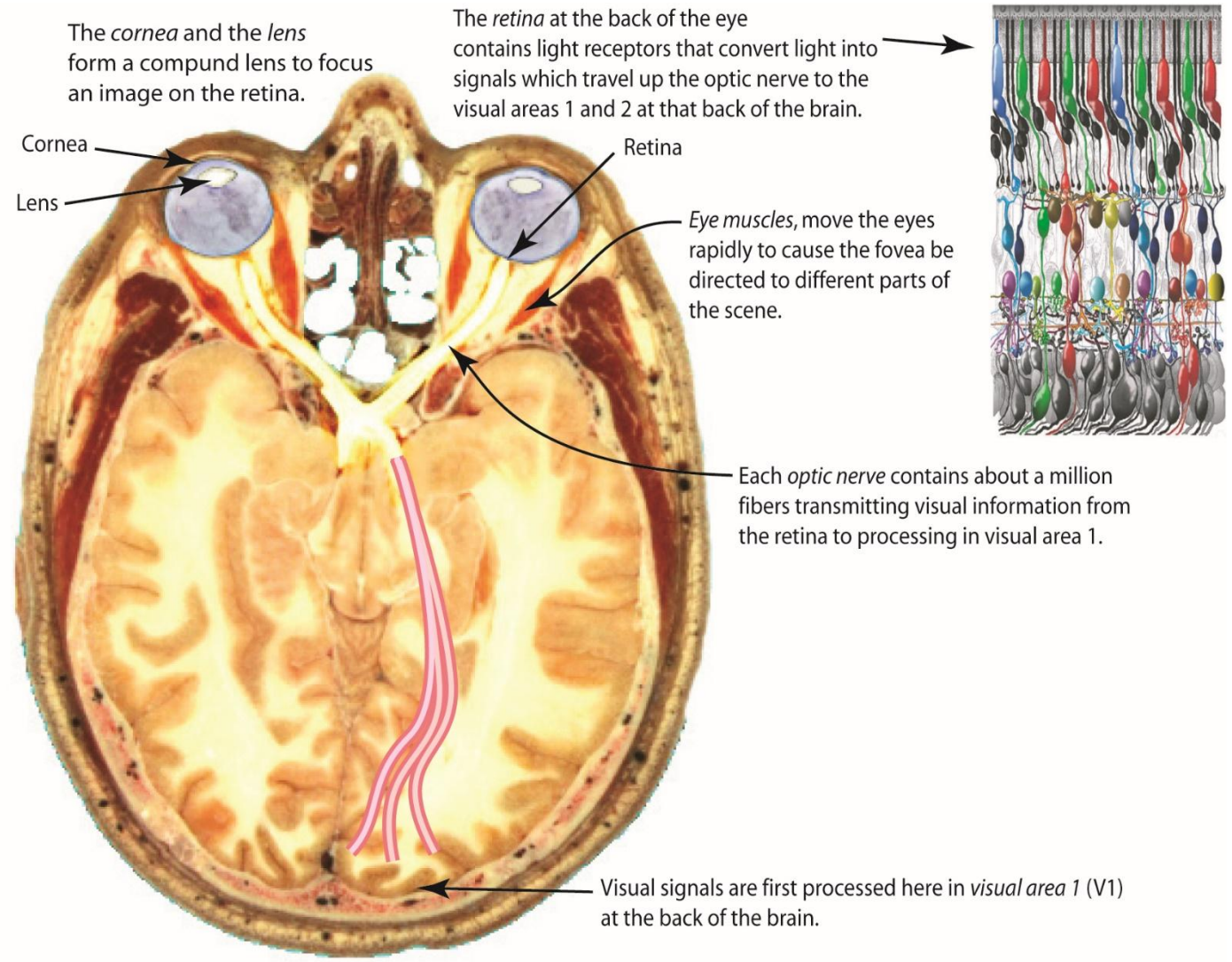
The Visual System
Our Eye





[Ware08]

The Visual System
Connection to Brain



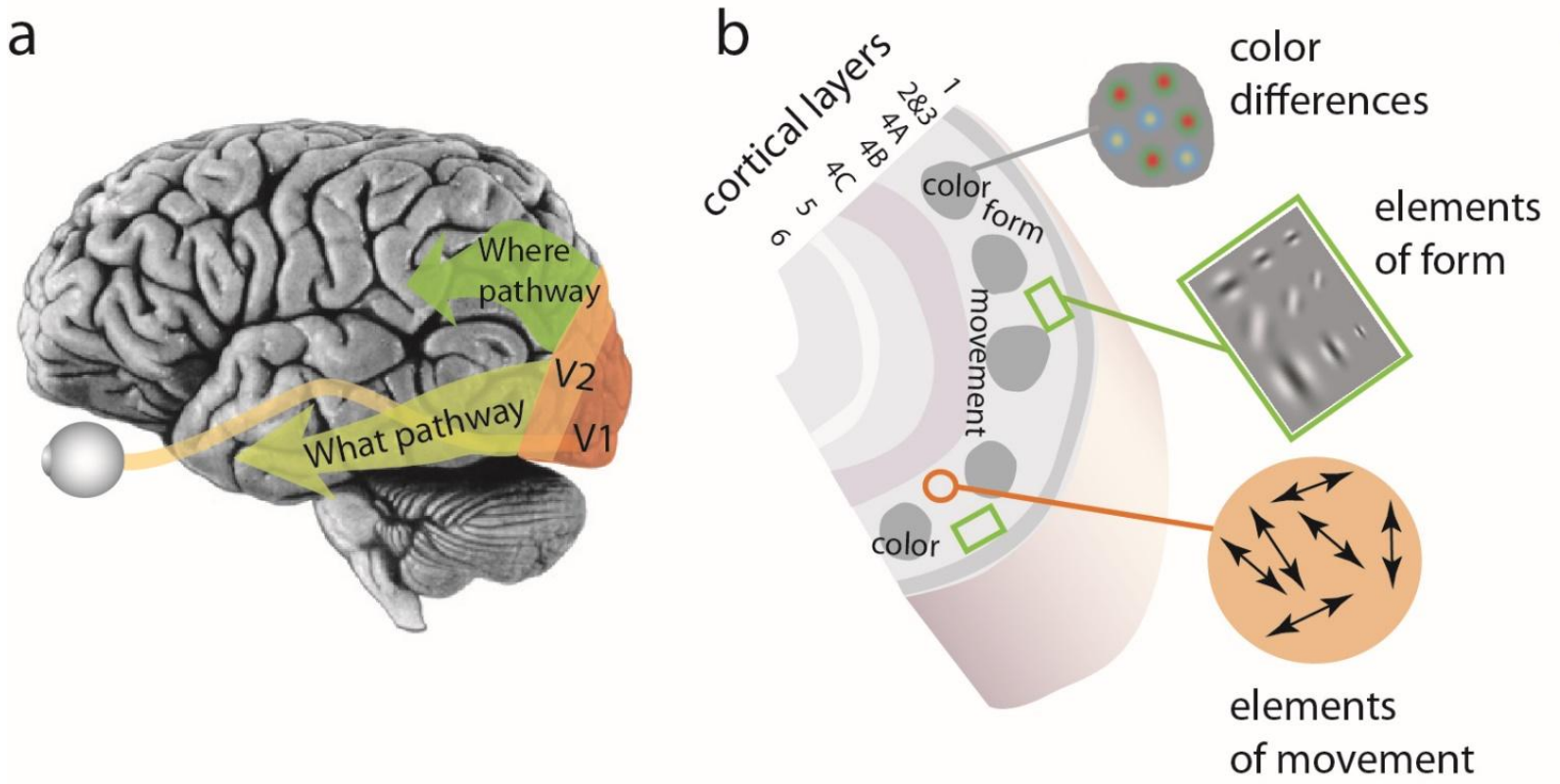
[Ware08]

Perception and Cognition

(3) Cognitive Processing & Pattern Recognition

TUD

Cognitive Processing
Pathways and the Primary Visual Cortex (V1)



Adapted from Livingston and Hubel [Ware08]

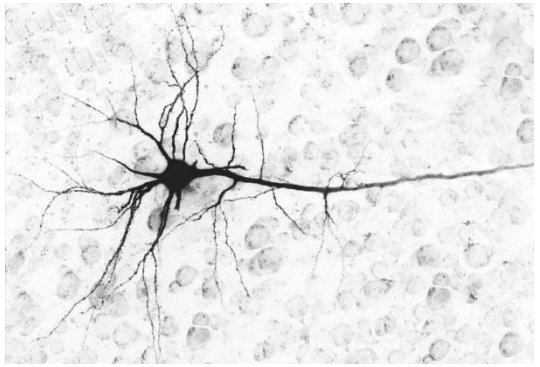
Pattern Processing
The “What” Pathway

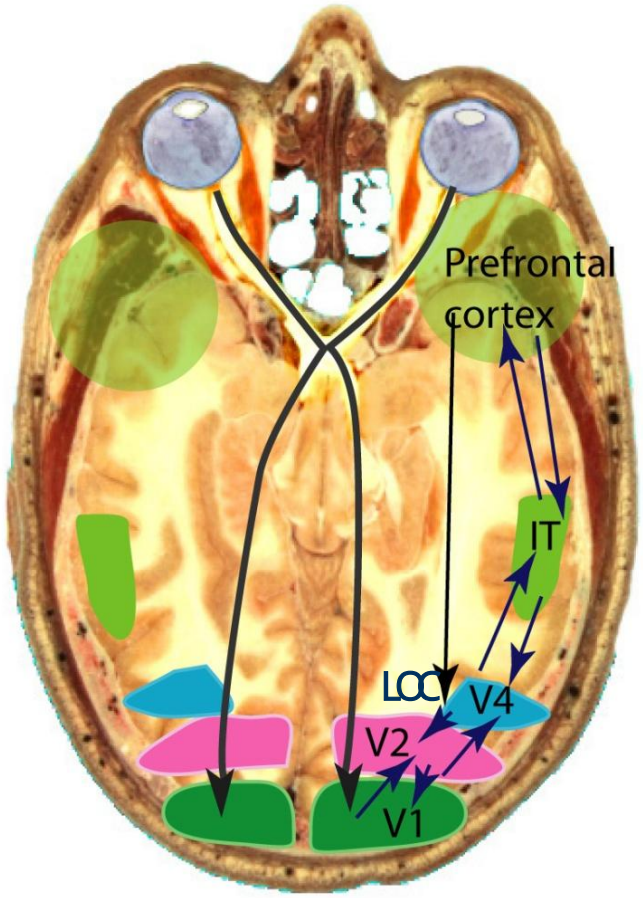
The “What” pathway has the function of identifying objects

Objects, any location →

Patterns of patterns →

Simple features specific locations →

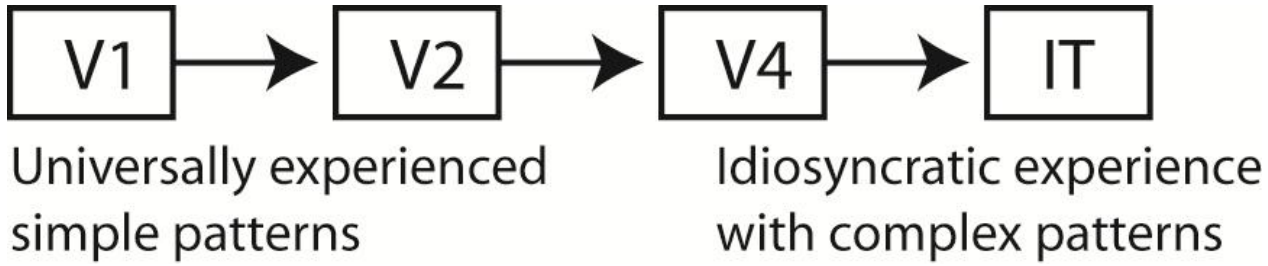




[Ware08]

Patterns
The “What” Pathway

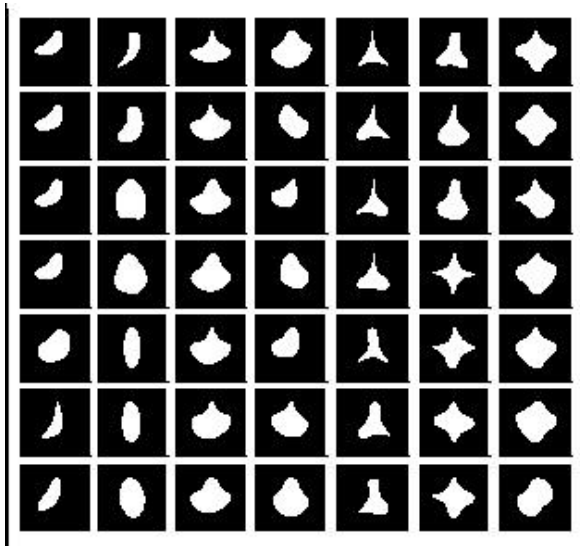
Feature hierarchy (learned at the top)



Contours and Regions (formed on the fly)





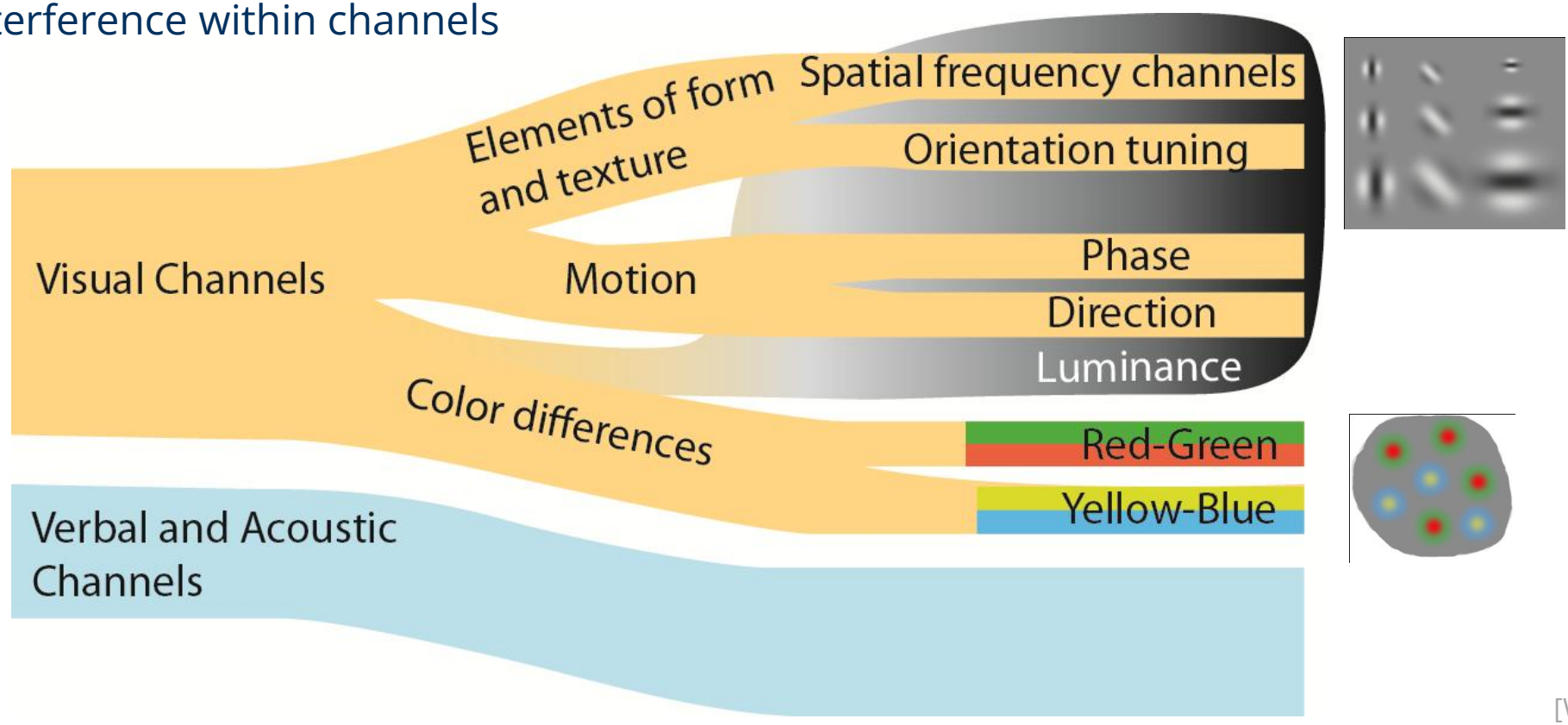


[Ware08]

Cognitive Processing
Channel Theory

Channels have sub-channels

- Visual separation between channels
- Visual interference within channels



[Ware08]

Cognitive Processing
Channel Theory – Concrete Example (tested in experiment)

Color sequence

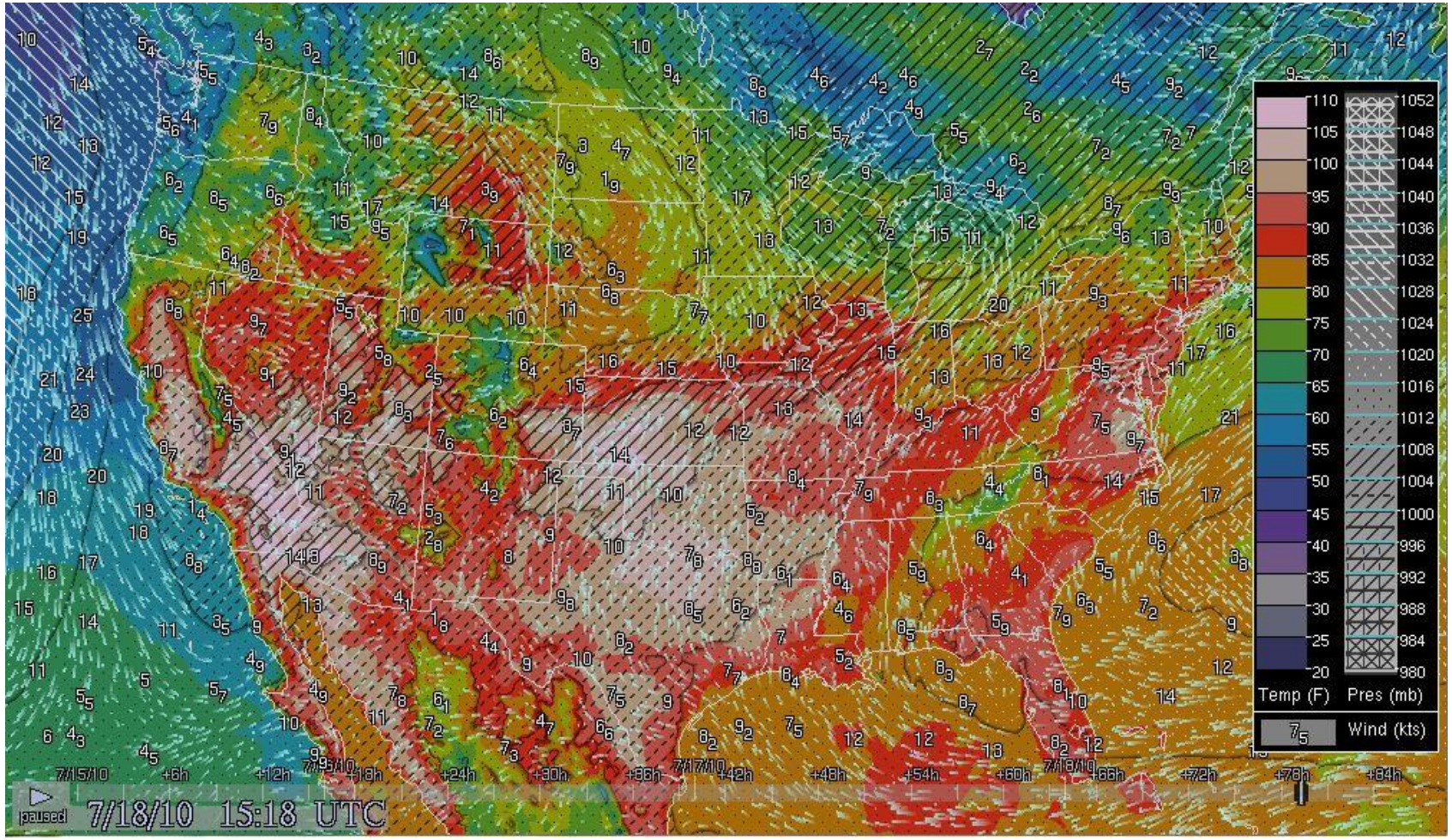
- for temperature

Texture sequence + Contours

- for pressure

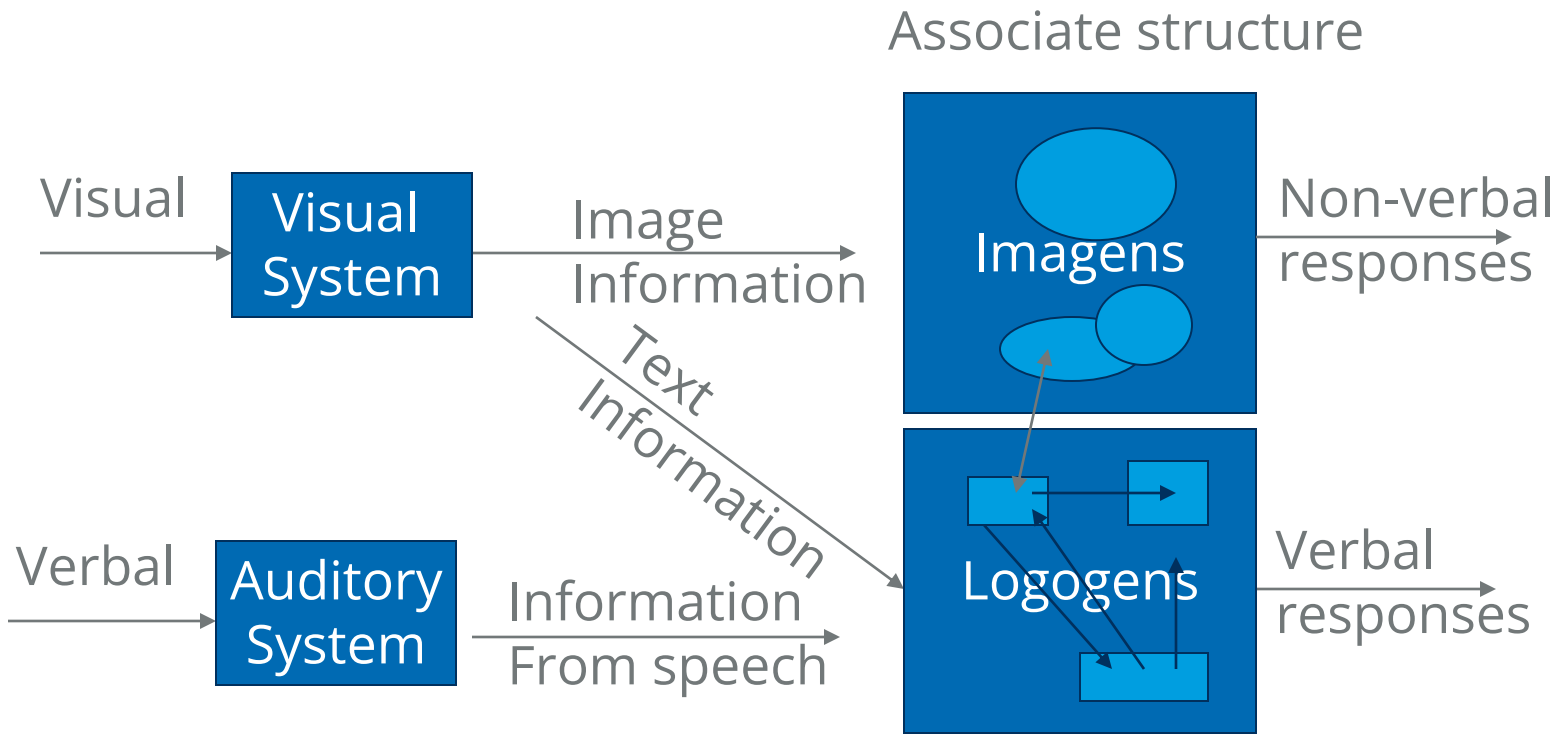
Motion:
animated streamlets

- wind speed
- Orientation
- + numbers



[Ware08]

Cognitive Processing
Pavio's dual coding theory

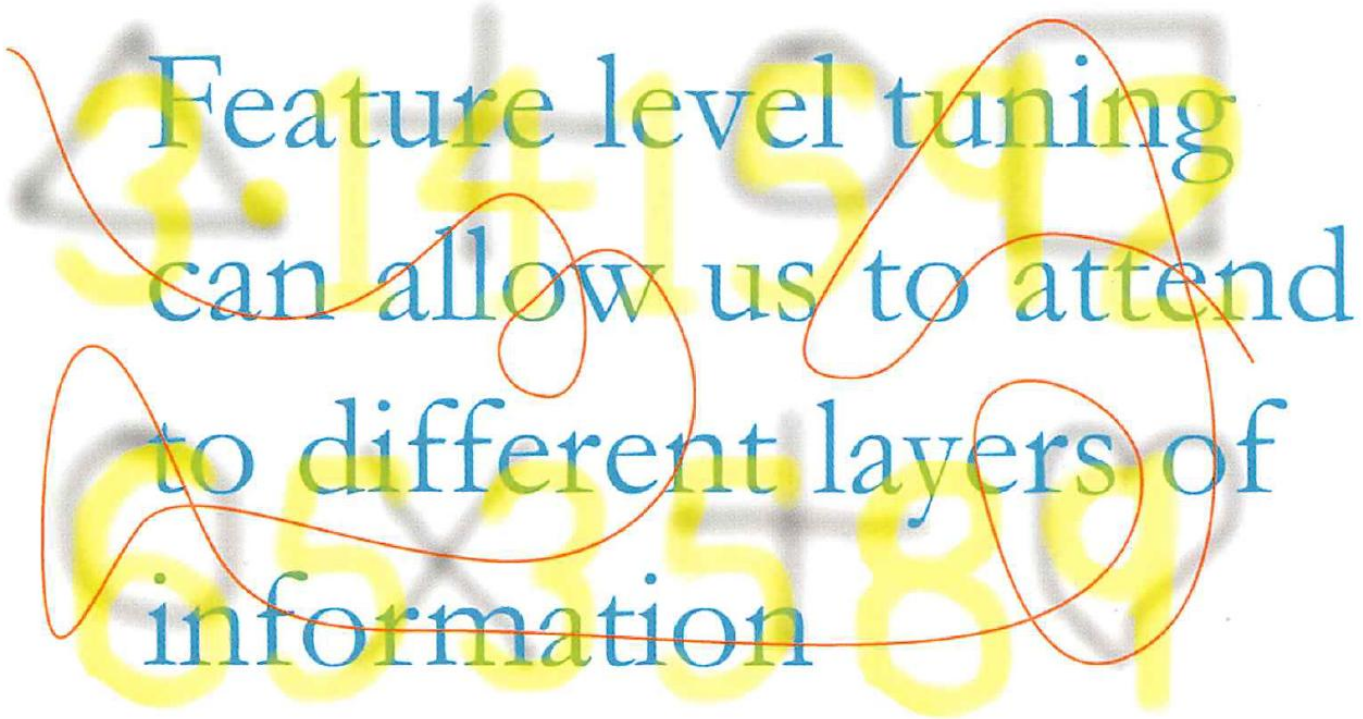


[Ware08]

Patterns
Attentional tuning

Attentional tuning operates at the feature level (not the level of patterns)

However, since patterns are made up of features, we can choose to attend to particular patterns if the basic features in the patterns are different.



[Ware08]

Perception and Cognition

(4) Preattentive Processing

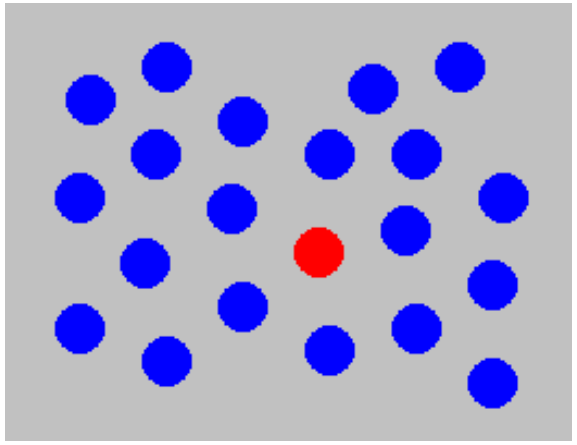
TUD

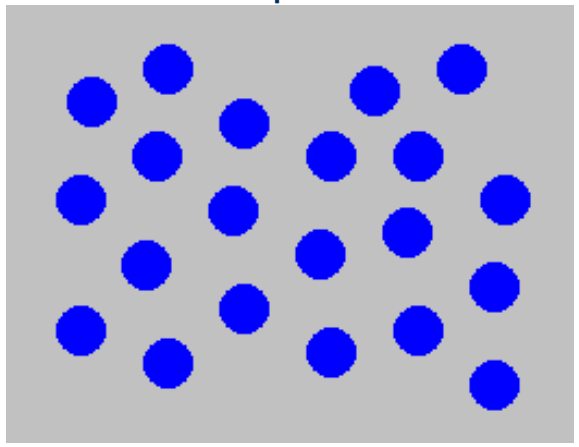
Preattentive Processing
Introductory Examples

Experiment of pop-out effects by Anne Treisman in 1988

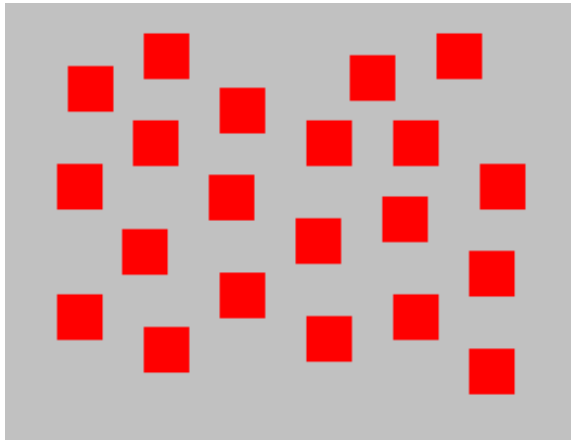
Observer can quickly and exactly determine, whether the target (red dot) is present or not.

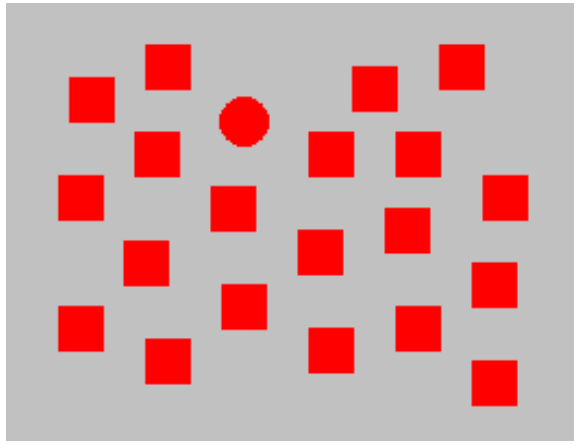
- Color allows the distinction:





- Form allows the distinction:





[Healey07]

Preattentive Processing
Basic Principle of pop-up effects

A limited set of visual properties are processed preattentively

- i.e., without need for focusing attention

This is important for designing visualizations

- What can be perceived immediately?
- Which properties are good discriminators?
- What can mislead viewers?

< 200 - 250ms qualifies as pre-attentive (slightly misleading, rather pop-up effects/attentional tuning)

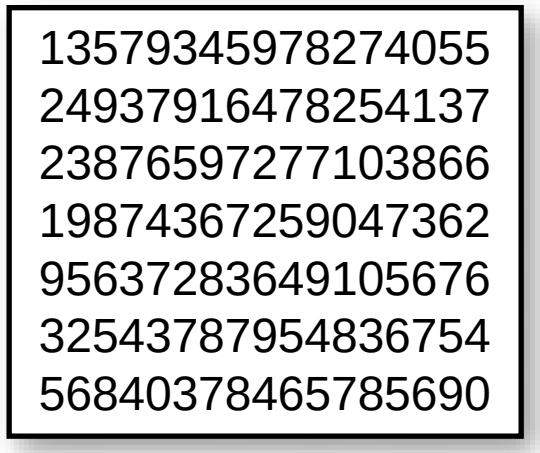
- Eye movements take at least 200ms
- Yet certain processing can be done very quickly, implying low-level processing in parallel

If a decision takes a fixed amount of time regardless of the number of distractors,
it is considered to be preattentive.

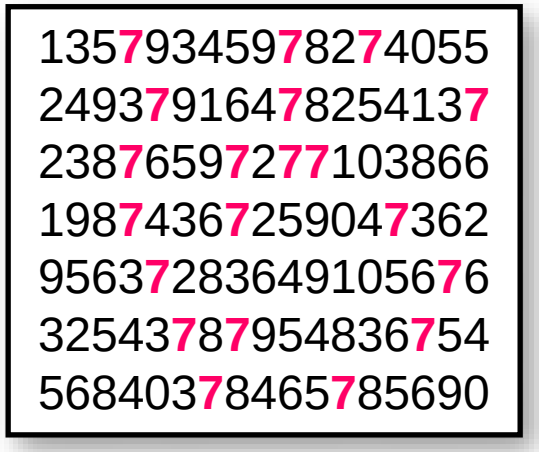
[Hearst08]

Preattentive Processing
Example and Test

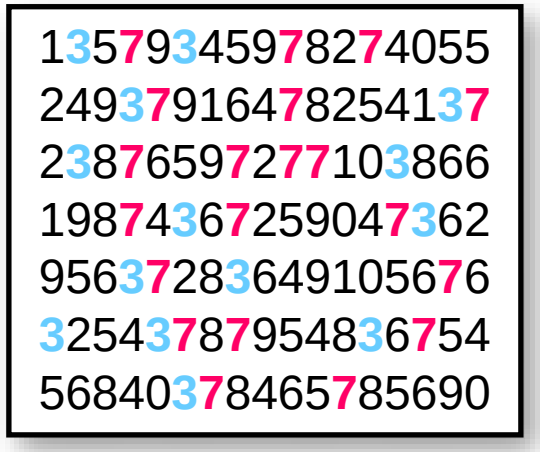
Count the Sevens!



Time proportional to number of digits



Time proportional to number of characters “7”

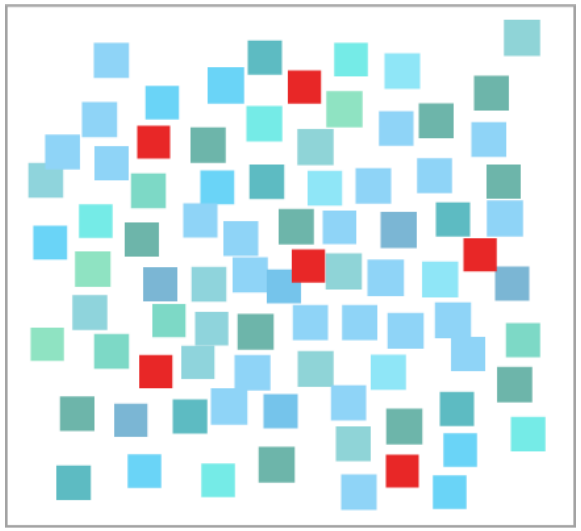


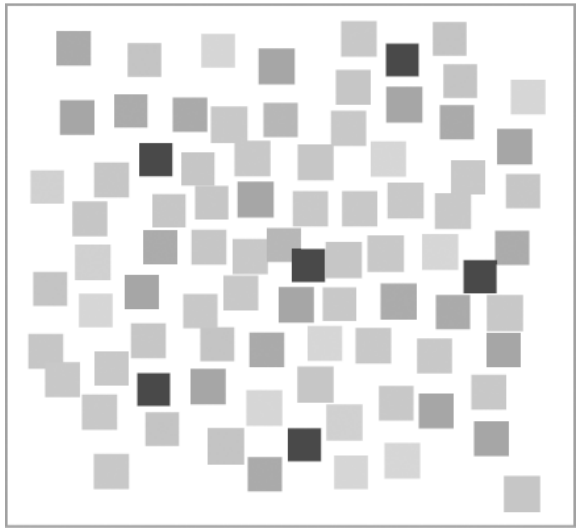
Both numbers 3 and 7 can be processed preattentively

[Hearst08]

Preattentive Processing
Contrast Creates Pop-out

- Hue and lightness | Lightness only





Pop-out

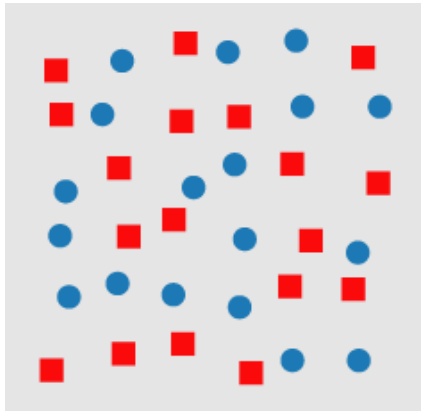
- Typically, 5-6 distinct values simultaneously
- Up to 9 under controlled conditions

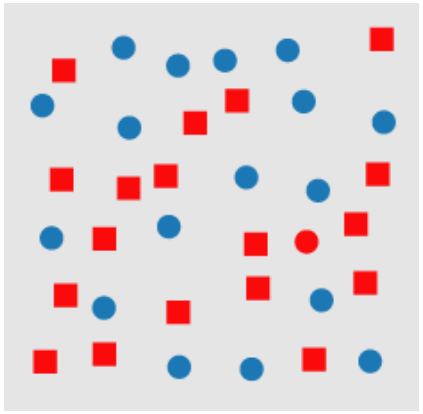
Distinguishable

- 20 easily for reasonable sized stimuli
- More if in a controlled context
- Usually need a legend

[Hearst08]

Preattentive Processing
Conjunction of Features





Viewer cannot rapidly and accurately determine whether the target (e.g. red circle) is present or absent when target has two or more features, each of which are present in the distractors. Viewer must search sequentially.

Demo: https://www.testable.org/project/public/c4e792a9-9524-4282-a954-dcd3e08ed49b

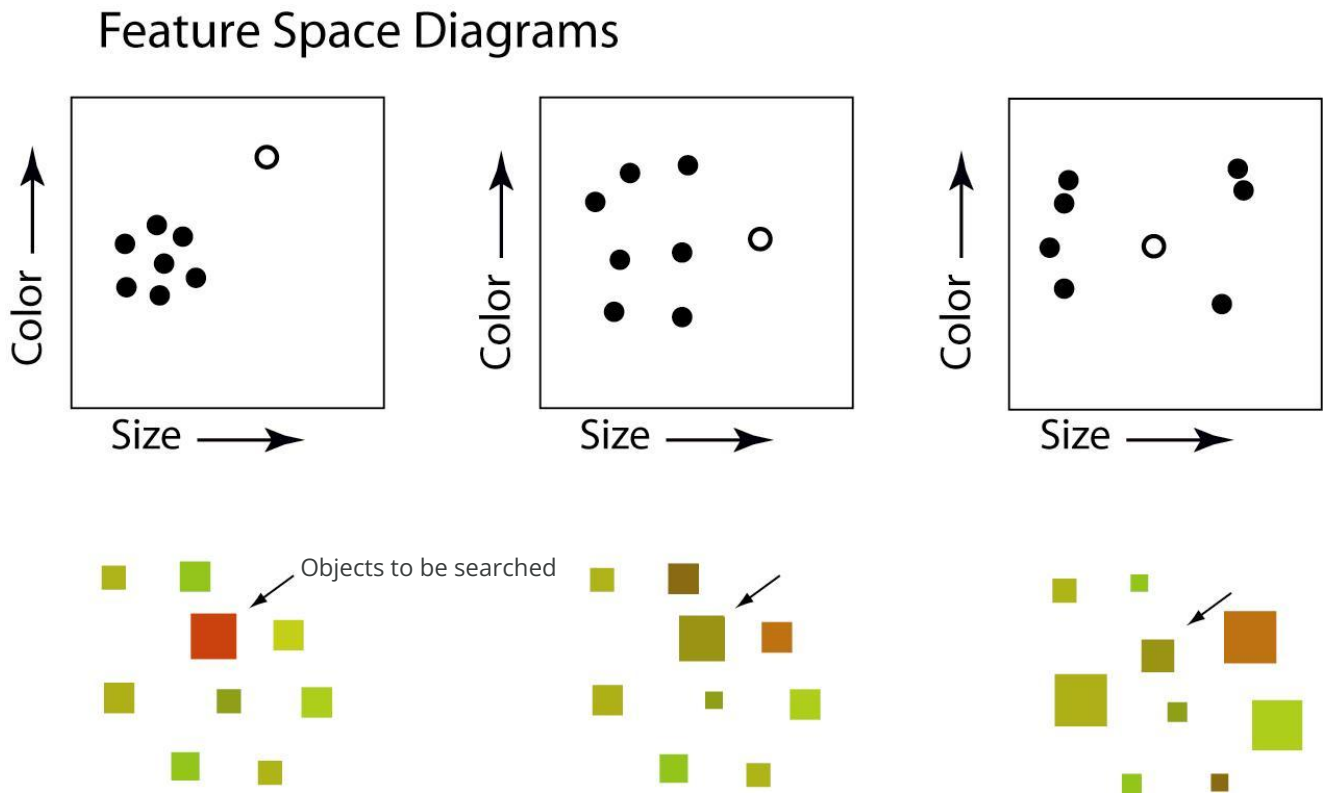
Preattentive Conjunctions

- Most of them are not preattentive!
- Exceptions:
  - Position/motion + (shape | color ...)
  - In general: spatial location and some aspect of form

[Healey07]

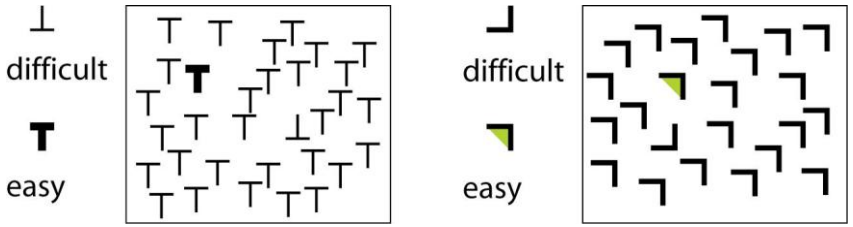
Preattentive Processing
Conjunction of Features

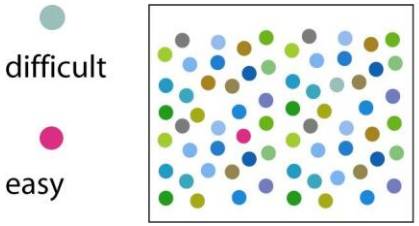
Redundant Coding

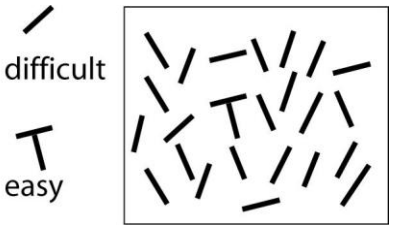


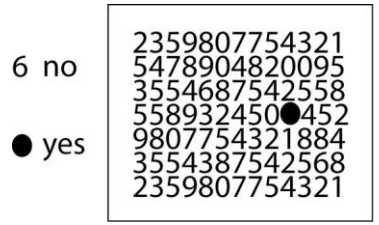
Preattentive Processing
Preattentive Features

But different levels of difficulty!

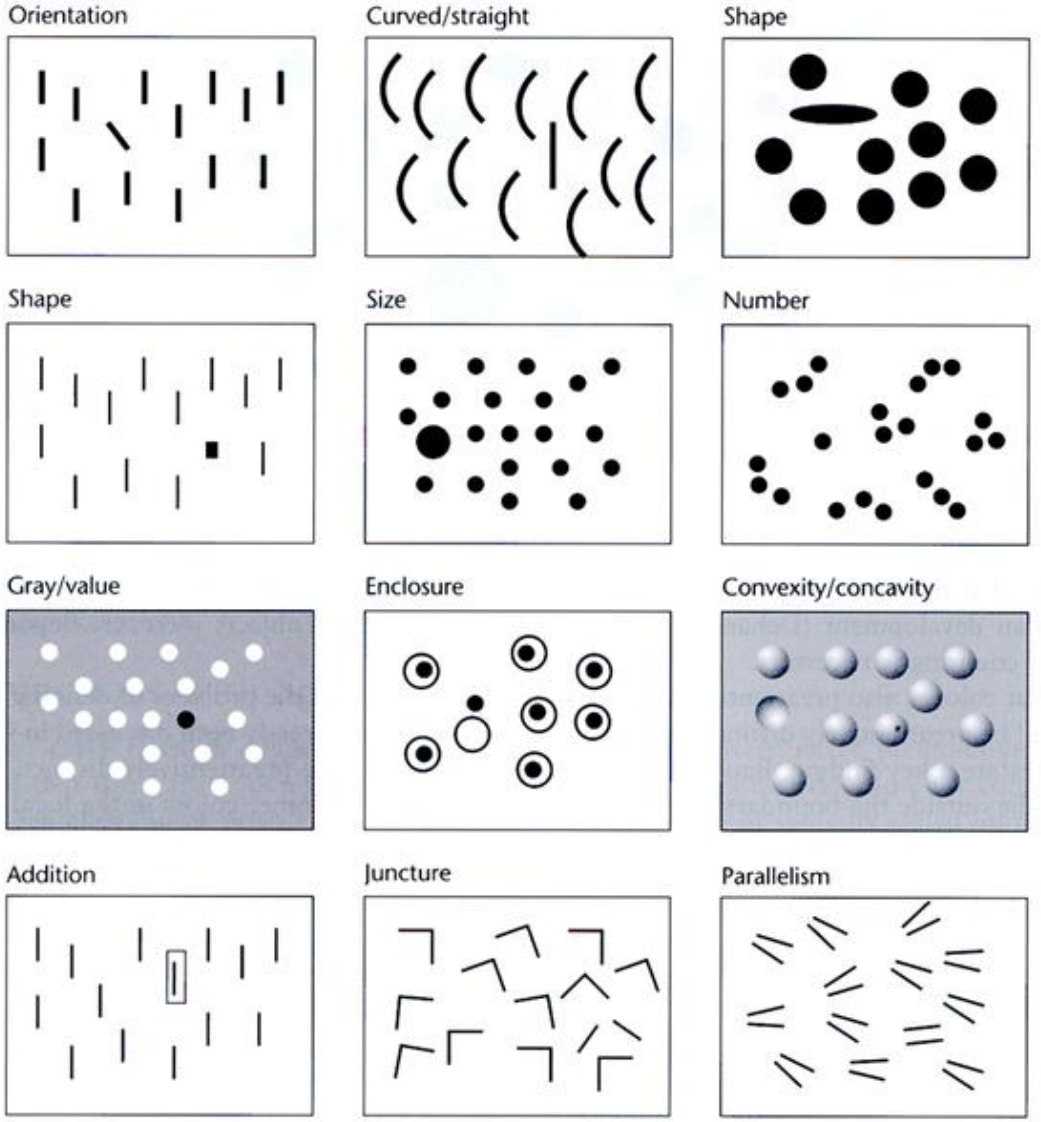












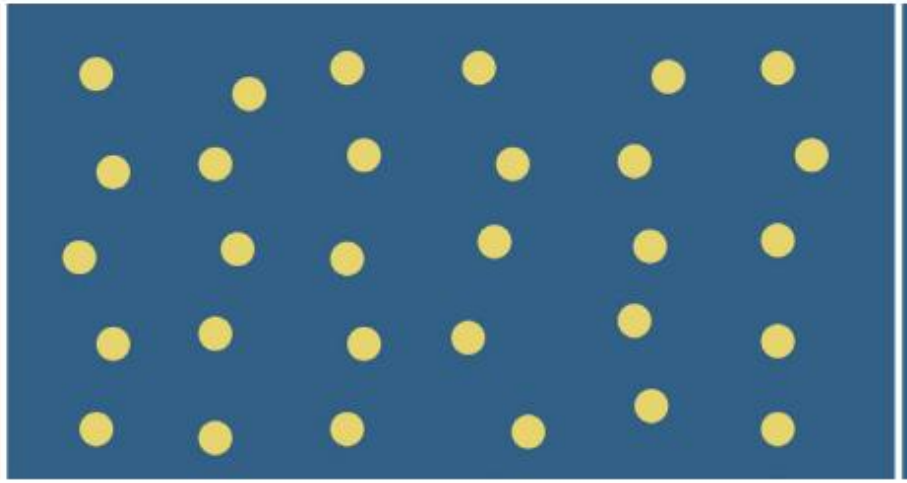
Preattentive Processing
Preattentive Features

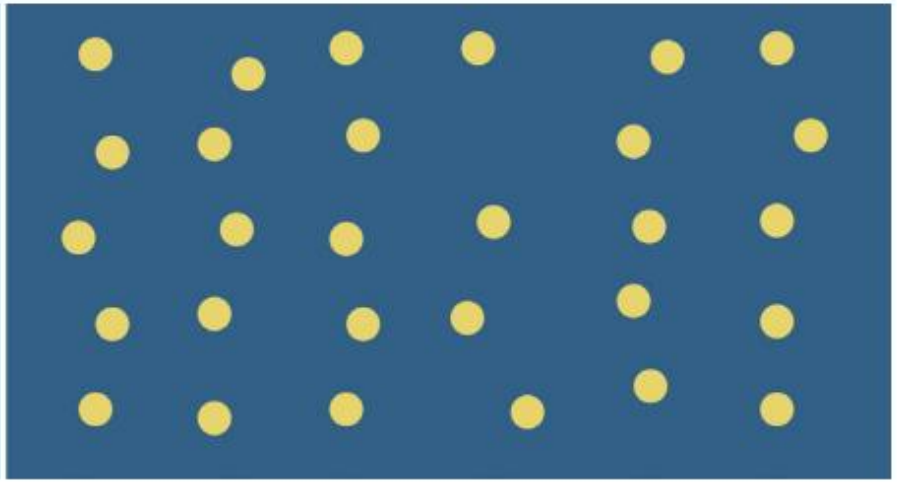
Research example from 2018: Deadeye https://ieeexplore.ieee.org/document/8440097

Presentation of two slightly different images, target object rendered for one eye only.

Does not alter visible properties of the target object – reduces risk of misinterpretation!







Perception and Cognition

(5) Gestalt Principles

TUD

Gestalt Principles
Overview

(Frankfurt) “Schule der Gestaltpsychologie” (from ca. 1912 on)

- Max Wertheimer, Kurt Koffka (1935), Wolfgang Kohler
- Also Gestaltgesetze, Gestalt Laws, Gestalt School
- Principles of pattern and form recognition

Forms and patterns are more than the stimuli being used to generate them

- Why are patterns emerging?
- What are the conditions for this?

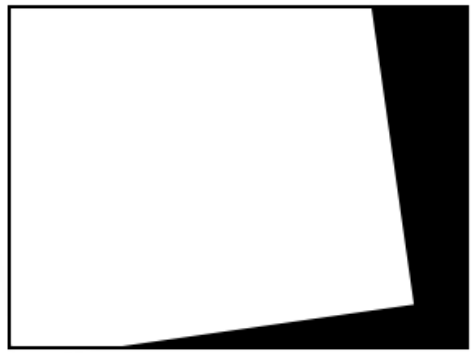
1. Prinzip der Figur-Grund-Trennung (Figure and Ground)
2. Prinzip der Ähnlichkeit (Similarity)
3. Prinzip der Nähe (Proximity)
4. Prinzip der Symmetrie (Symmetry)
5. Prinzip der Konvexität (Geschlossenheit, Closure)
6. Prinzip der stetigen Fortsetzung (Continuity)
7. Prinzip der Relativen Größe (Relative Size)

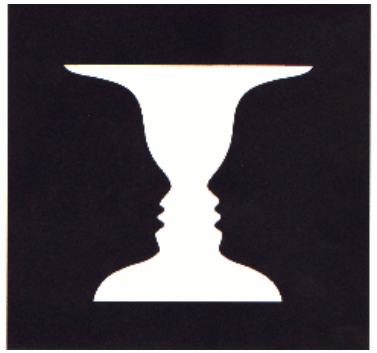
Connectedness (a later addition)
Common fate (a later addition)

Gestalt Principles
Figure / Ground

Basic Principle

- Visual system distinguishes foreground (objects, figures) and background
- Important for object identification
- Which object attracts attention?
- Elements contribute like
  - relative sizes, white spaces, closed contours, symmetry







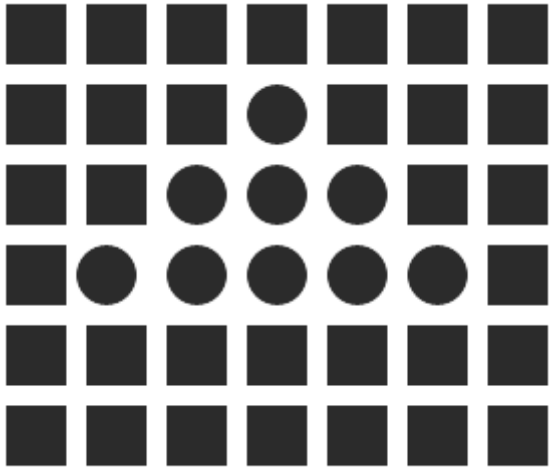


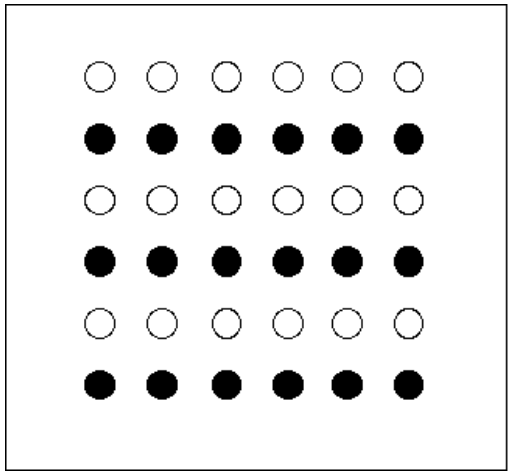


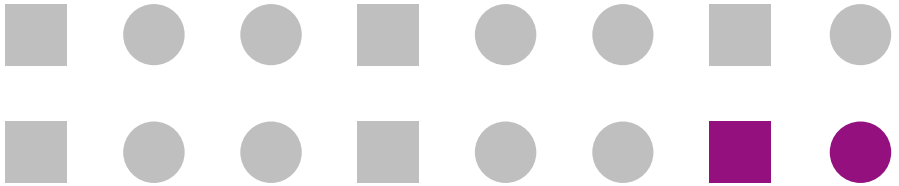
- Why does Rubin‘s Vase challenge us?
  - Competing recognition processes
  - Emphasis on foreground object eases it

Gestalt Principles
Similarity

- Similar objects will be perceived as belonging together
  - Size, form, color
  - Color groups stronger than form





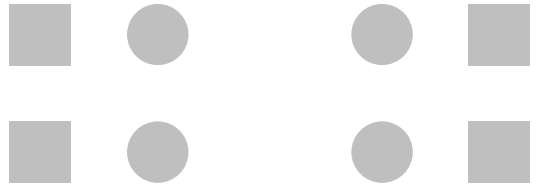


Gestalt Principles
Proximity

- Close objects will be perceived as belonging together
- Proximity stronger than Similarity
- One of the strongest principles
  - Foundational for expressing relations between objects
  - Important for clustering (principle of Spatial Concentration [Slocum ‘83])

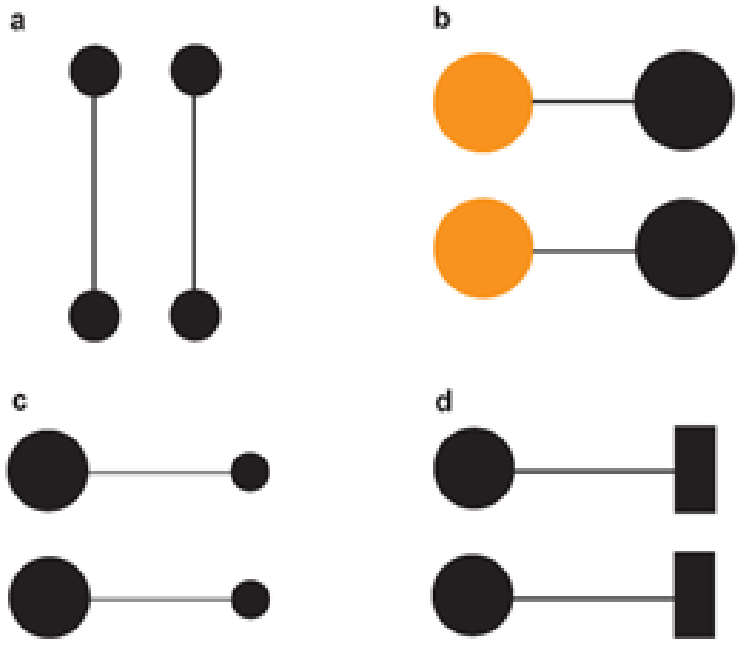






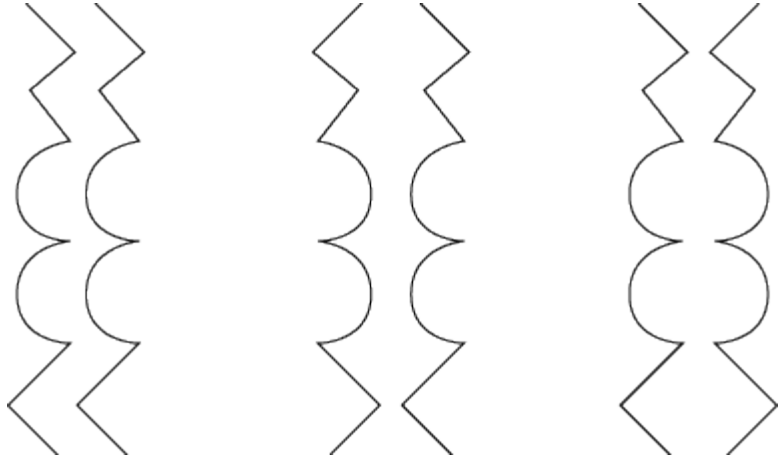
Gestalt Principles
Connectedness

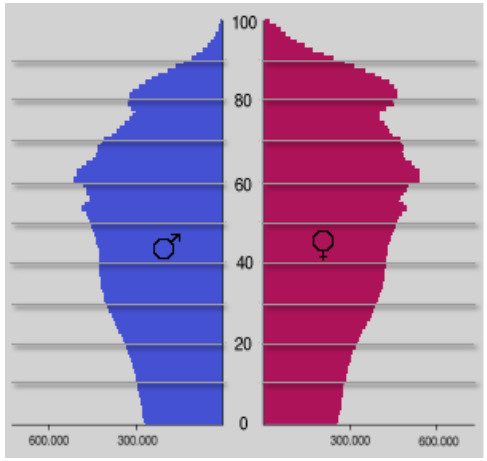
- Addition by Palmer & Rock (1994)
- Even stronger than Similarity (size, color, form) and Proximity
- Can even override other principles
- Inhomogeneous elements can be grouped by lines around them or background

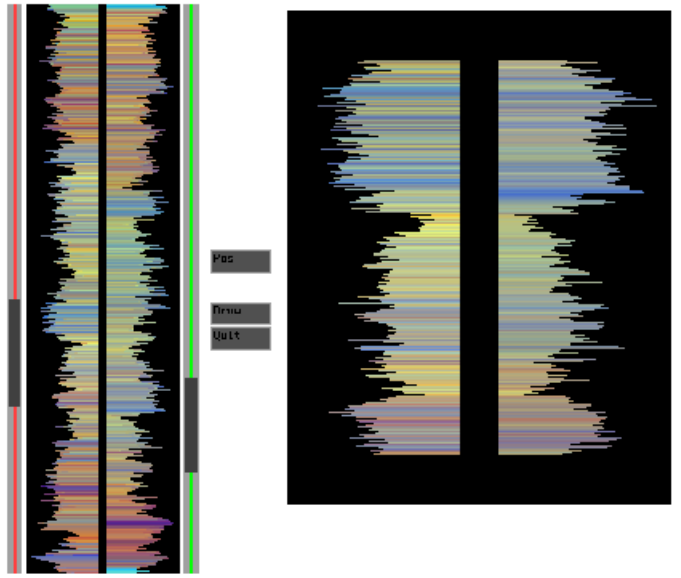


Gestalt Principles
Symmetry

- Connection of similar objects along symmetric axis
- Bilateral symmetry creates strong figural impression
- Examples: age pyramids, time series data



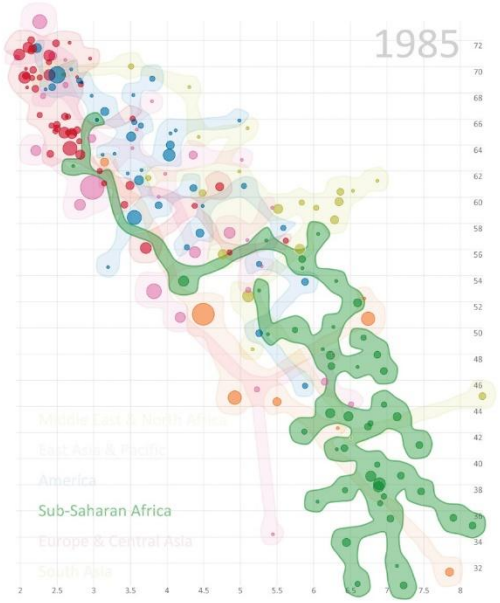




Gestalt Principles
Closure (and Common Region)

Geschlossenheit

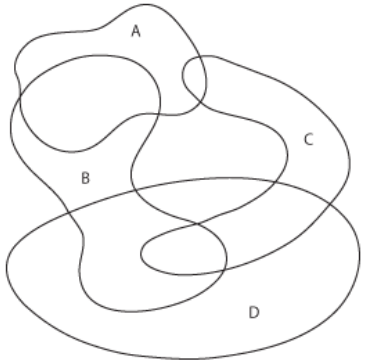
- Closed contours to show set relationship
- Form does not need to be complete to be perceived as a whole
- Humans add remaining bits cognitively
- Convex forms are preferred
- Application: grouping information

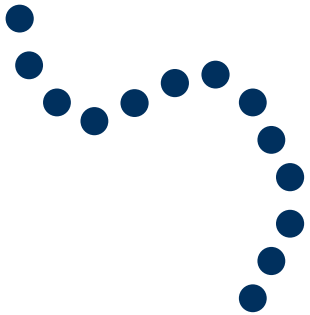


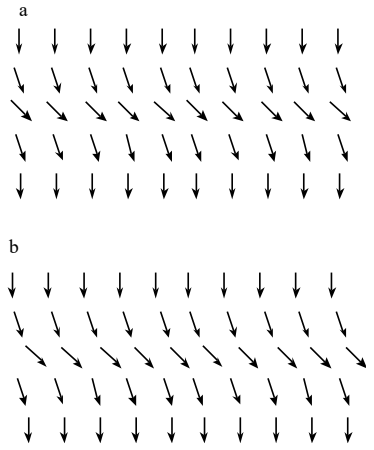
Example: Collins Bubble Sets





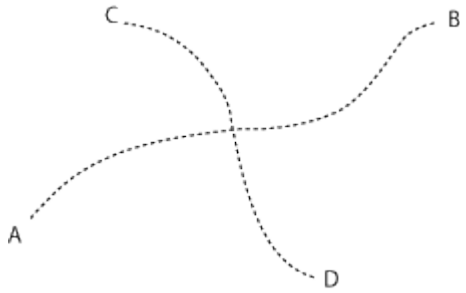






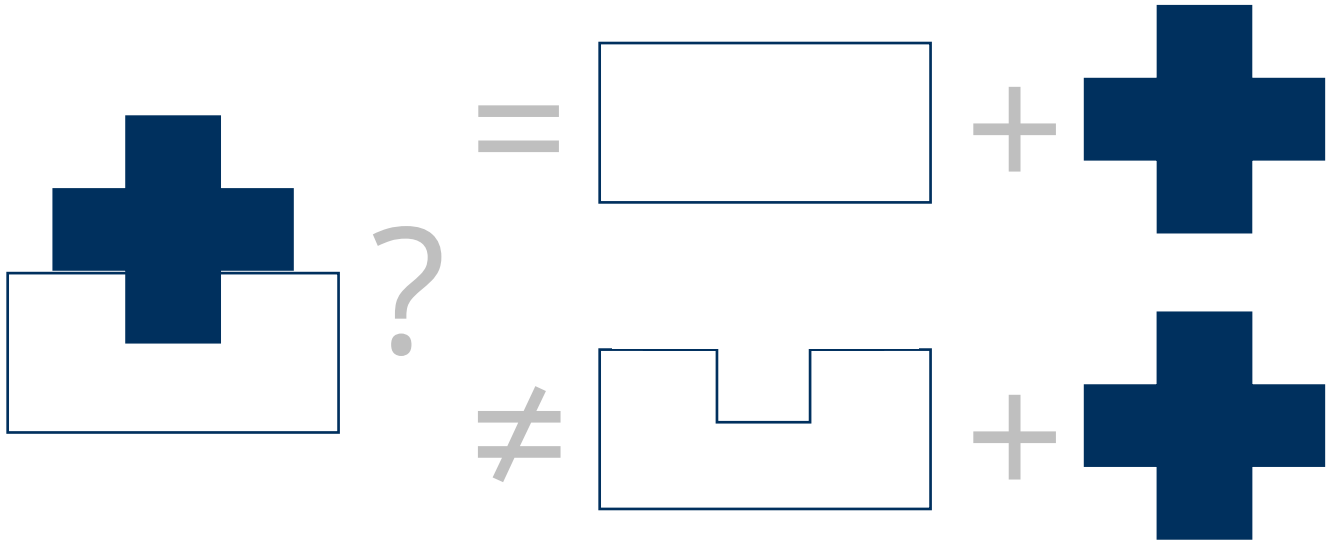
Example: Vector Fields

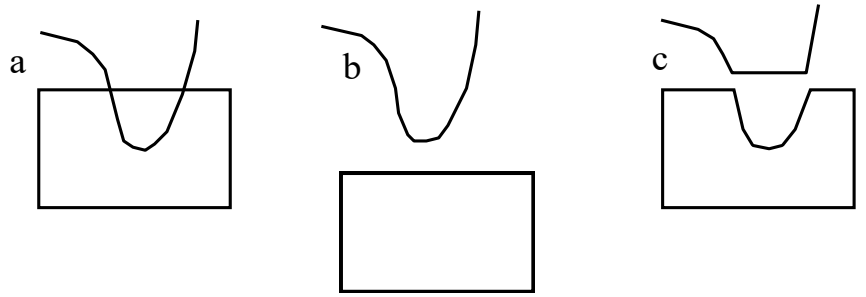
Gestalt Principles
Continuity



Also: Smooth Gradient

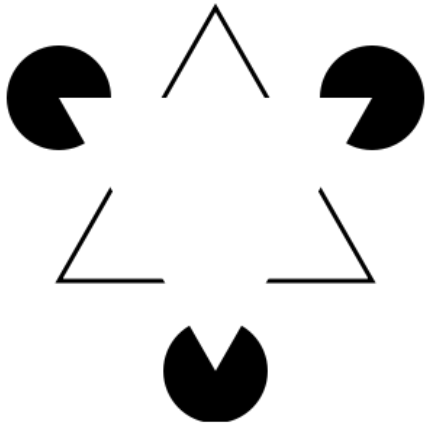
- Visual entities tend to be smooth and continuous
- Contours are mentally continued so that they do not
  - change direction abruptly
  - Introduce new forms

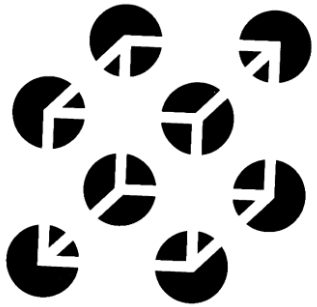


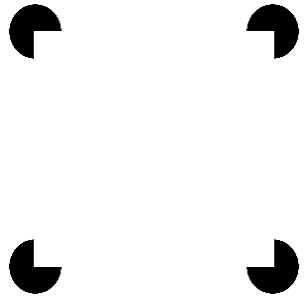


Gestalt Principles
Simplicity (Prägnanz, gute Gestalt)

- Are many geometric elements thinkable, people choose the simplest
  - Especially basic geometric shapes
  - E.g., Kanizsas triangle
- These forms are also an example for the principles of Closure



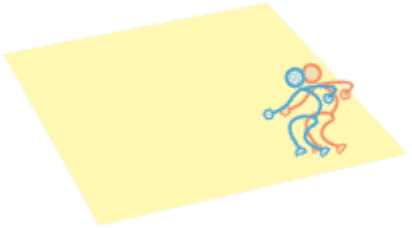


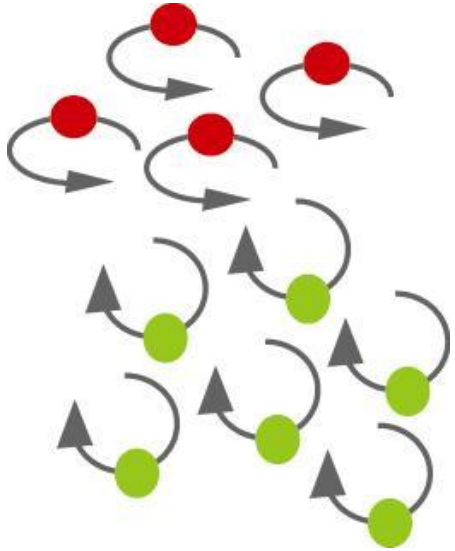


Gestalt Principles
Common Fate

- Similarity perceived in dynamic animations
  - Objects moving in the same direction appear to be connected



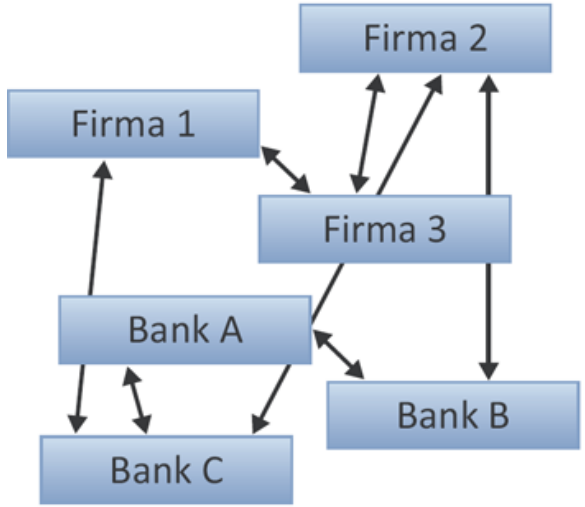


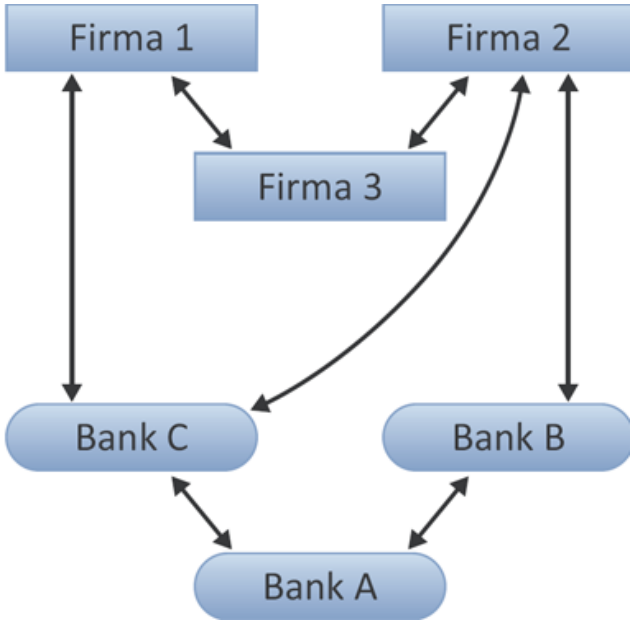


Gestalt Principles
Example

Example

- Why is the right diagram perceived faster and better?





- Both:
  - Connectivity
- Only right diagram
  - Proximity, Continuity, Similarity, Symmetry

Gestalt Principles
Conclusion

What we can learn

- Gestalt Principles are useful as design guidelines
- Patterns should be present in luminance
- Patterns should have the appropriate size
- Motion is under-researched,
  but evidence suggest its power
- Simple motion coding can be used to express
  - communication, causality, urgency, happiness? (Braitenberg)

Perception and Cognition

(6) Perception of Color and Contrast

TUD

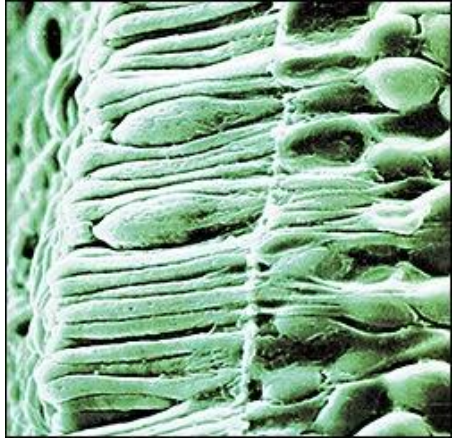
Color Perception
Physiology: Receptors

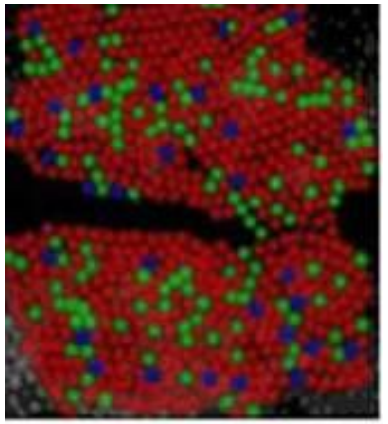
Rods

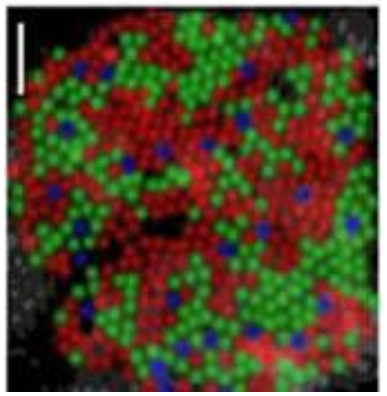
- active at low light levels (night vision)
- only one wavelength sensitivity function
- 120 million rod receptors

Cones

- active at normal light levels
- three types: sensitivity functions peaks at different wavelengths
  - “red” long-wavelength-sensitive cones (L)
  - “green” middle-wavelength-sensitive cones (M)
  - “blue” short-wavelength-sensitive cones (S) (far less!)
- 6.5 million cone receptors
- Focused in the center of vision (fovea)

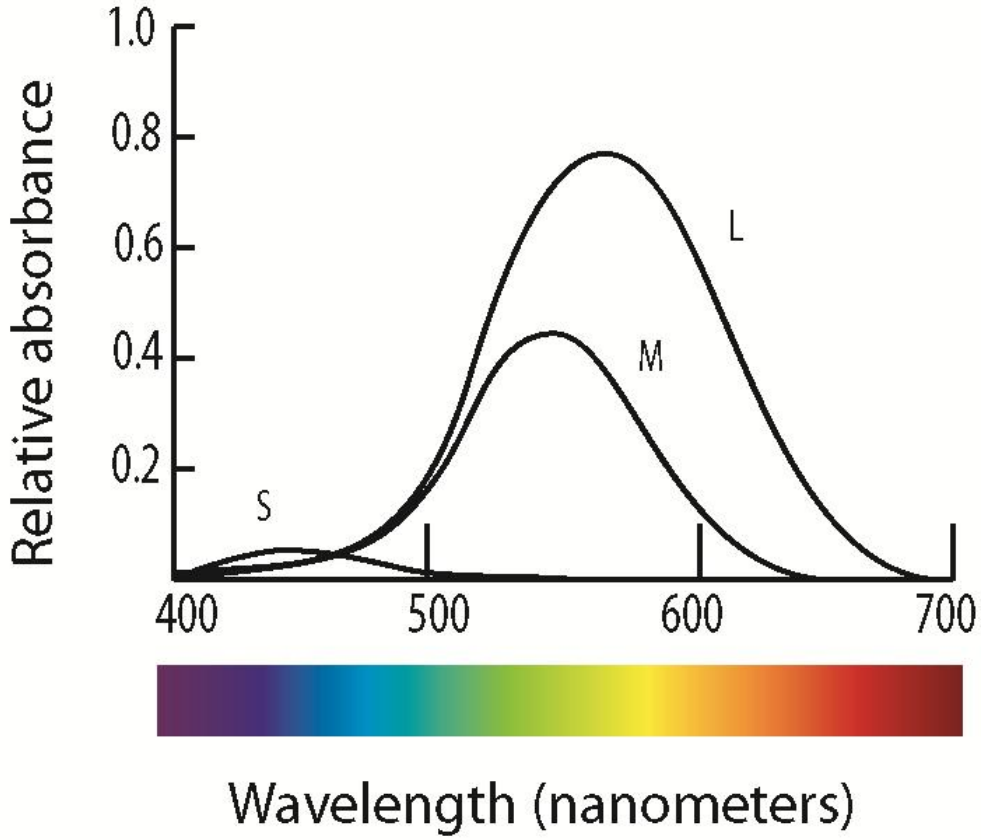






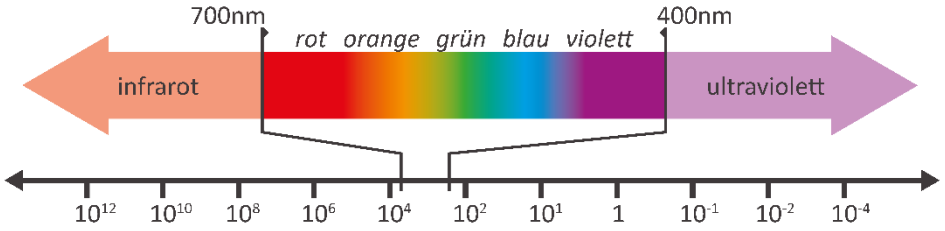
Color Perception
Cone Sensitivity Functions

Cone Sensitivity Functions are the basis of color vision and measurement



Color Perception

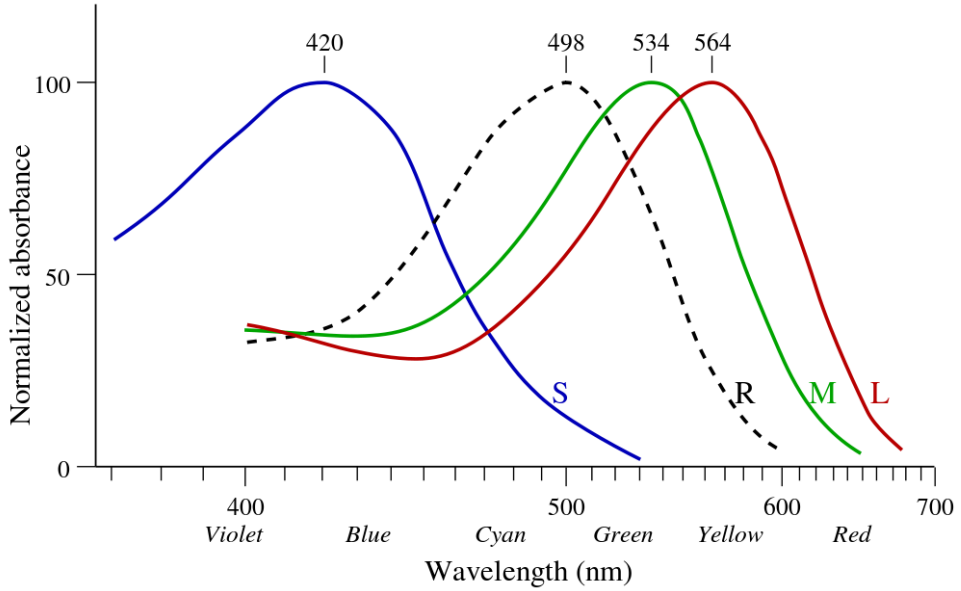
Human eye perceives light with wavelengths between 380 and 780 nm

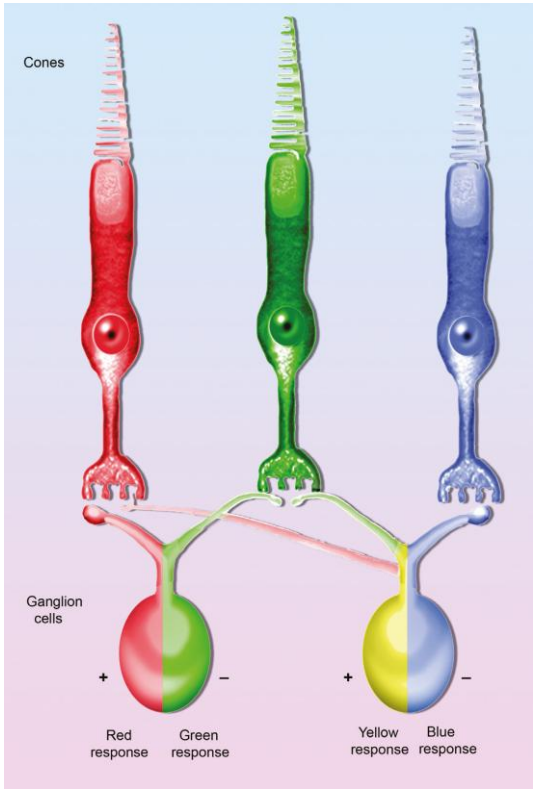


Natural Color Space:

Additive mixture of SML

Trichromatic vs. Color opponent theory





Color Perception
Color-opponent Channel Properties

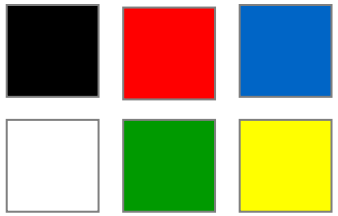
Channels: red-green, yellow-blue, black-white

Simultaneous Contrast





Unique Hues



Sensitivity to spatial detail, motion, depth, shapes

- Luminance channel: much greater capacity to convey detailed information than chromatic channels
- Luminance channel: conveys motion information much more effectively than chromatic channels
- Stereoscopic depth information done via the luminance channel
- Luminance channel can process shape-from-shading, whereas chromatic channels cannot

Color Perception
Color-opponent Channel Properties

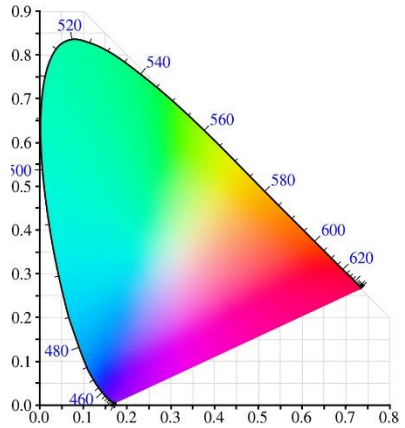
Luminance nonlinearity

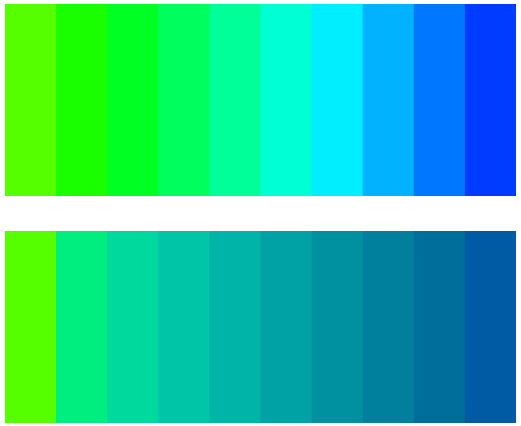
- We are more sensitive to dark gray differences than to light gray differences
- Also background dependent!
- Light background increases differences near white *(sharpening)*



Colors

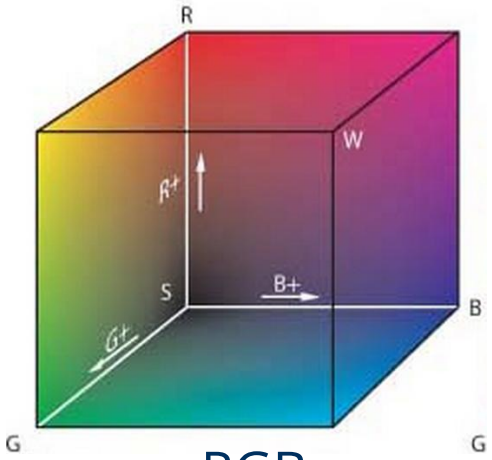
- Attention tuning possible (homogeneous background improves this)
- Different ways of laying out the 3D color space (resulting from three types of cones) possible:
  - Not perceptually uniform color spaces (e.g., RGB)
  - Perceptually uniform color spaces (e.g., L*a*b, HSLuv)



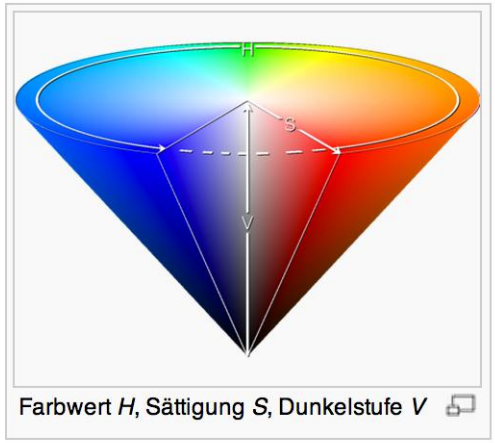


Color Perception
Color Spaces

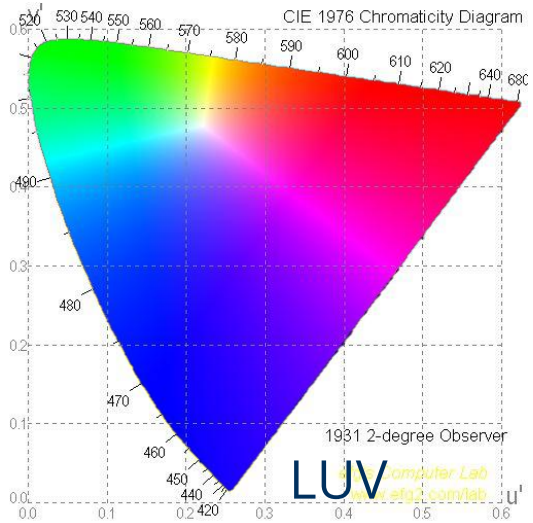
Different color spaces with different characteristics and purposes:



RGB



HSV



LUV

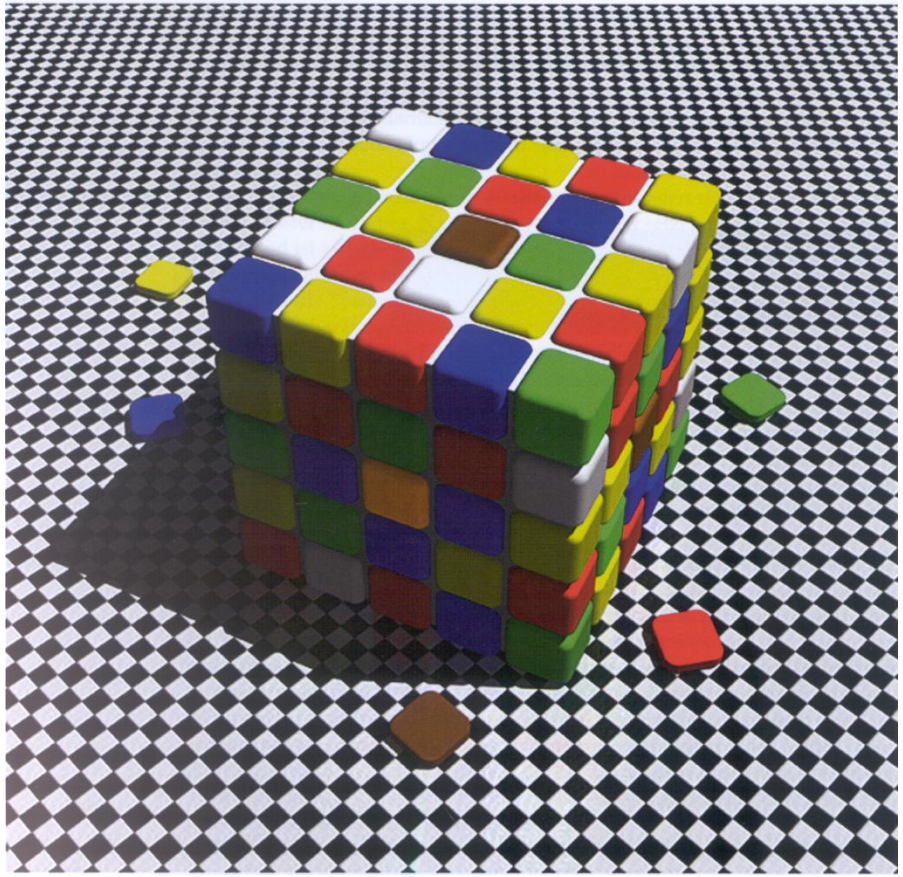
RGB: technical color space (additive, linear) used by displays, does not include entire spectrum

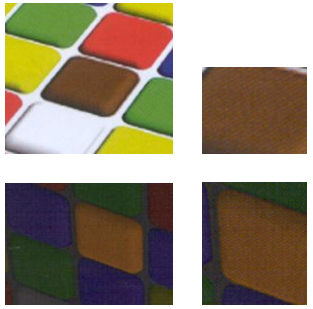
HSV: more natural, perception-oriented parametrization in Hue, Saturation and Value (brightness)

CIE 1976 L*, u*, v*: transformation of the CIE XYZ color space with perceptual uniformity

COLOR PERCEPTION

Color constancy. Brown (on top) and orange (in the shadow) are clearly distinguished. The actual light intensity and wavelength are the same. © Dale Purves





COLOR PERCEPTION

Color perception depends on the visual context and on the viewer. Examples:

- Color appears less saturated on a dark background compared to a light background.
- Color contrast between a foreground object and background has an influence on the perceived size of an object.
- Some colors (bright, strongly saturated colors) seem to in front of others (closer to the observer).
- High frequent data (with same colors) appear less saturated.

Consequence: color perception must be evaluated in a specific situation to find out whether the intended effect is actually achieved.

COLOR PERCEPTION

Constant foreground color on a changing background

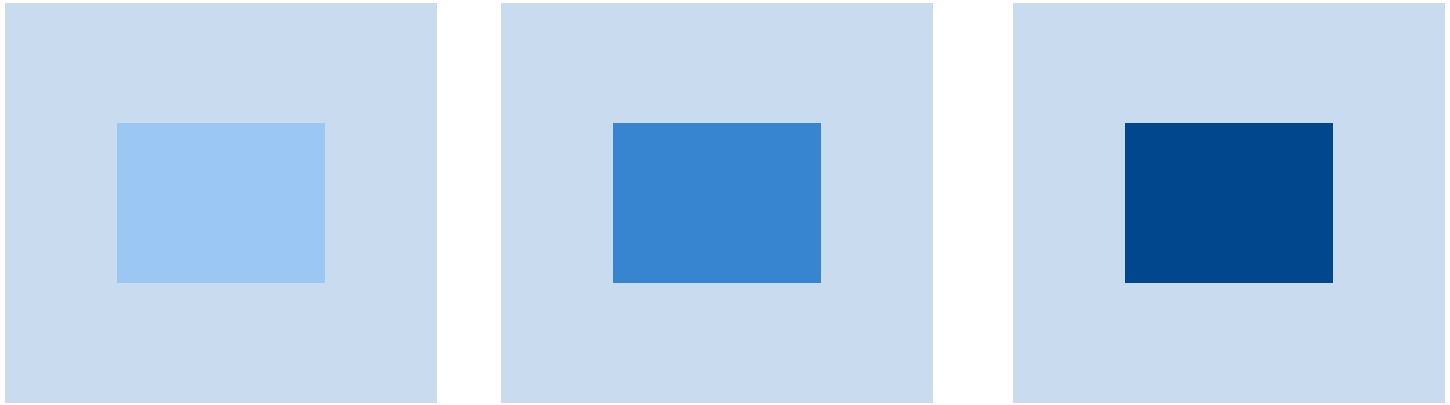
Rule 1: Color appears less saturated on a dark background compared to a light background.



COLOR PERCEPTION

Constant background color with a varying foreground

Rule 2: Color contrast between a foreground object and background has an influence on the perceived size of the object (higher contrast = bigger size).



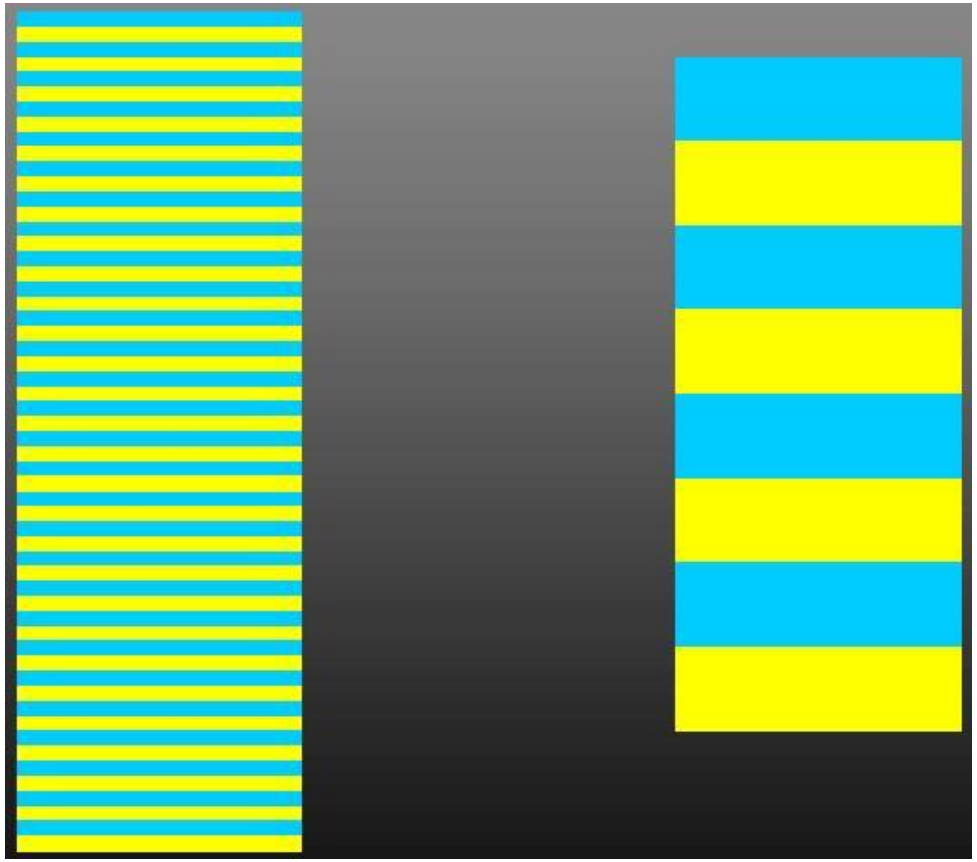
COLOR PERCEPTION

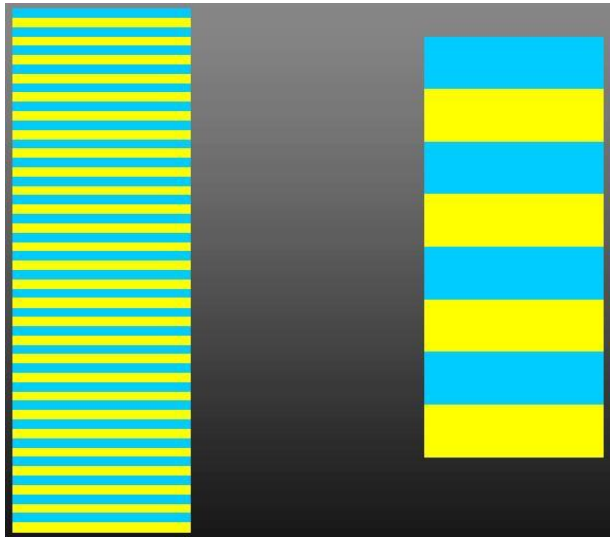
Rule 3: Some colors (bright, strongly saturated colors) seem to be in front of others / closer to observer.

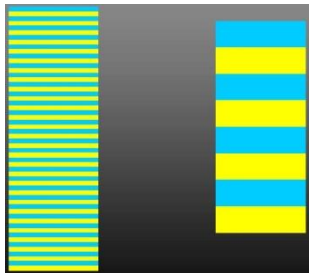


COLOR PERCEPTION

Rule 4: High frequent data appear less saturated (when same color combination is used).





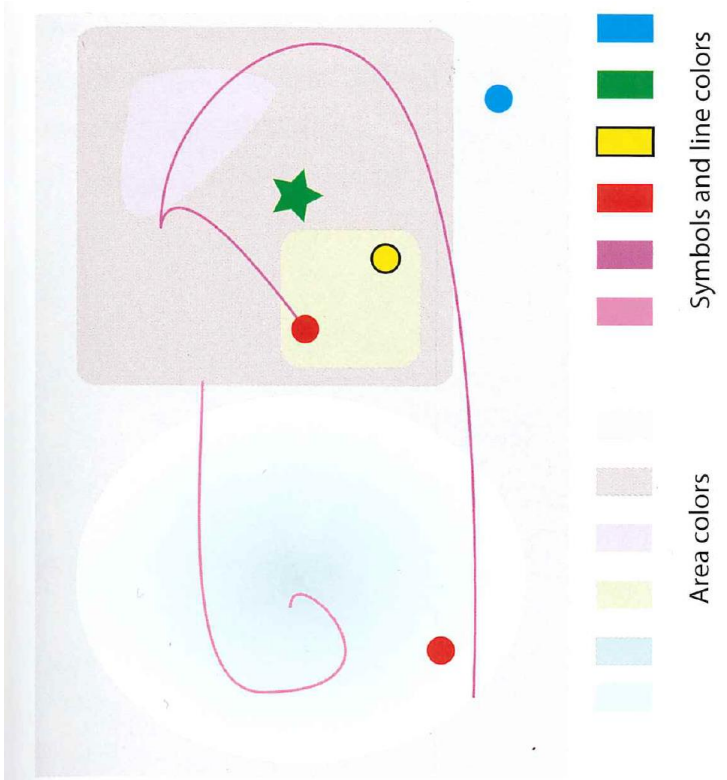


Bernhard Preim - 2008

COLOR PERCEPTION

(Some) Principles
for the use of color:

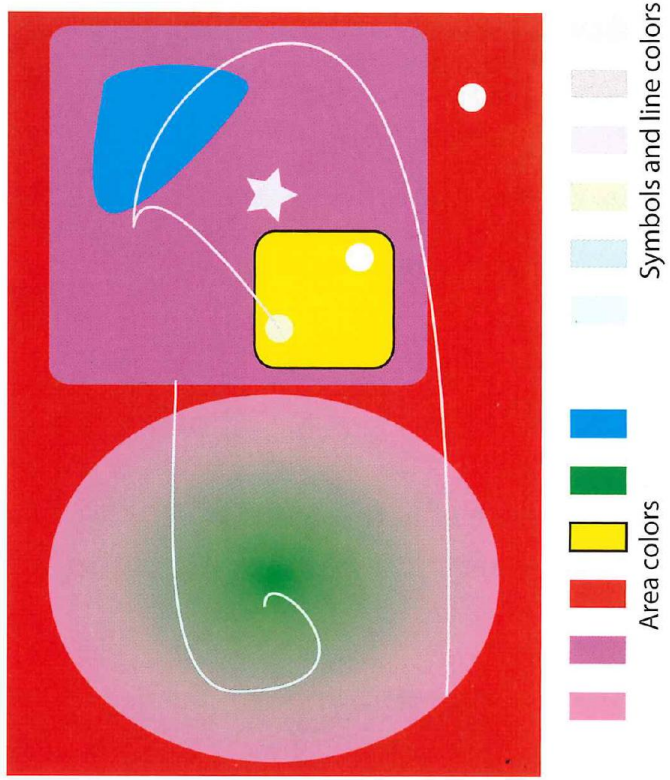
- Use more saturated colors when color coding small symbols, data marks, thin lines etc.
- Use less saturated colors when coding larger areas



GOOD. This example uses high saturation (strong colors) to code small areas such as symbols and lines. Larger background areas are all light and low saturation.

A black contrasting border is used to separate the yellow circle from its background because both have similar luminance.

[Ware08]



BAD. This example shows what happens when the colors used for small areas and large areas are switched. The color codes used for the symbols and lines are difficult to discriminate because of contrast effects.

Besides being completely ineffective, the second example would generally be regarded as unattractive.

COLOR PERCEPTION

(Some) Principles for the use of color:

- Economical, goal-directed use of color
- Use strongly saturated colors carefully (prefer pastel colors)
- Consider application-specific conventions concerning the meaning of colors (e.g. red – high temperature, blue – very cold)

In addition (see distribution of rods and cones):

- Do not use pure blue for text, thin lines or small shapes.
- Do not use blue for fast moving objects.
- To discriminate colors, it is insufficient to change the blue component only slightly.

Perception
Nonlinear Scaling

Psychophysics develops relationships between the magnitude of a stimulus Φ and the magnitude of the sensation Ψ.

Detection Threshold $\boldsymbol{\Phi_0}$: smallest stimulus that leads to perception

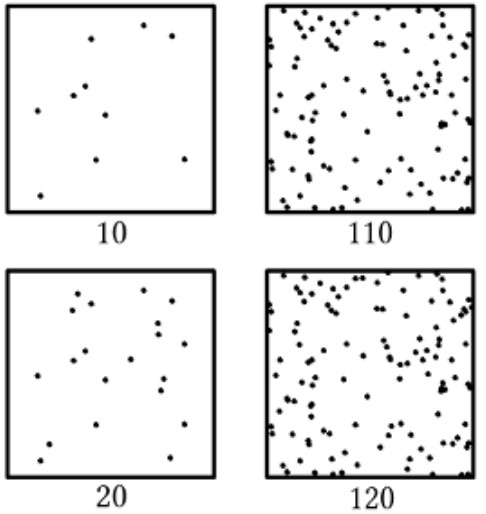
- Example: Brightness of darkest star that can be perceived on night sky
- Detection threshold is an *absolute* quantitative measure

Just-Noticeable Difference (JND) $\boldsymbol{\Delta\Phi}$: smallest change to the stimulus Φ that can be perceived.

- Example: smallest brightness difference of a point such that it can be perceived on grey background
- JND is a quantitative measure depending on the initial stimulus

Weber's Law: $\frac{\Delta\Phi}{\Phi+k} = \text{const}$ *the perceived change in stimuli is proportional to the initial stimuli*

Illustration ➔



From this, one can derive:

Fechner's Law: $\Psi = k \cdot \log \Phi$

- *Perceived brightness is proportional to logarithm of the actual intensity measured accurately* (only valid for medium intensity range)

Perception
Nonlinear Scaling

Stevens's Power Law

As a result of numerous user studies, S. S. Stevens published in 1957 an empirically found relationship between stimulus Φ and the magnitude of sensation Ψ:

$$\Psi = k \cdot \Phi^a$$

a is an exponent that depends on the type of stimulus and k is a scaling factor used only for normalization.

After taking the logarithm on both sides of the equation, a corresponds to the slope:

$$\log \Psi = a \cdot \log \Phi + \log k$$

Sensation magnitude is a power function of stimulus intensity

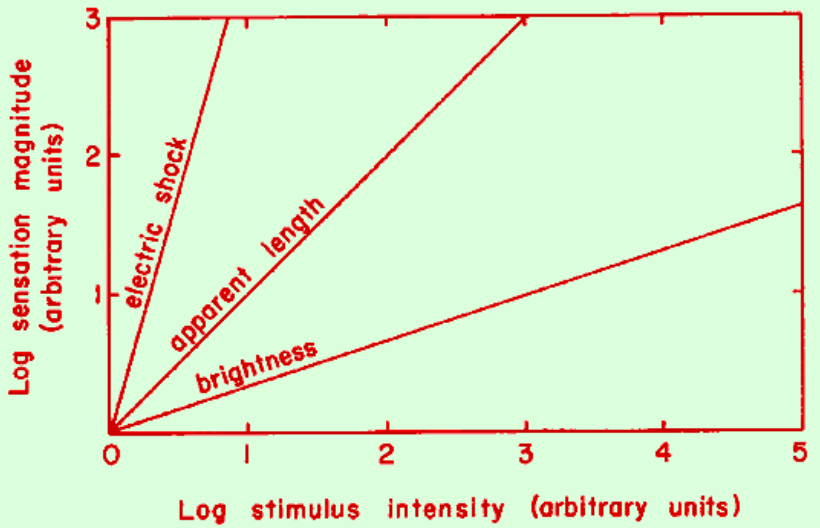


FIG. 8.3. Psychophysical magnitude functions for three perceptual continua. The linearity of the functions on double logarithmic coordinates indicates that sensation magnitude is a power function of stimulus intensity. The slope of the line corresponds to the exponent of the power function. The exponents for electric shock to the fingertips, line length, and the brightness of relatively large stimuli lasting about 1 sec are 3.5, 1.0, and .33, respectively.

Continuum	a	Stimulus condition
Brightness	0.33	5° target in dark
Brightness	0.5	Point source
Brightness	5	Brief flash
Brightness	1	Point source briefly flashed
Lightness	1.2	Reflectance of gray papers
Visual length	1	Projected line
Visual area	0.7	Projected square
Redness (saturation)	1.7	Red-gray mixture

Exponents of visual Stimuli

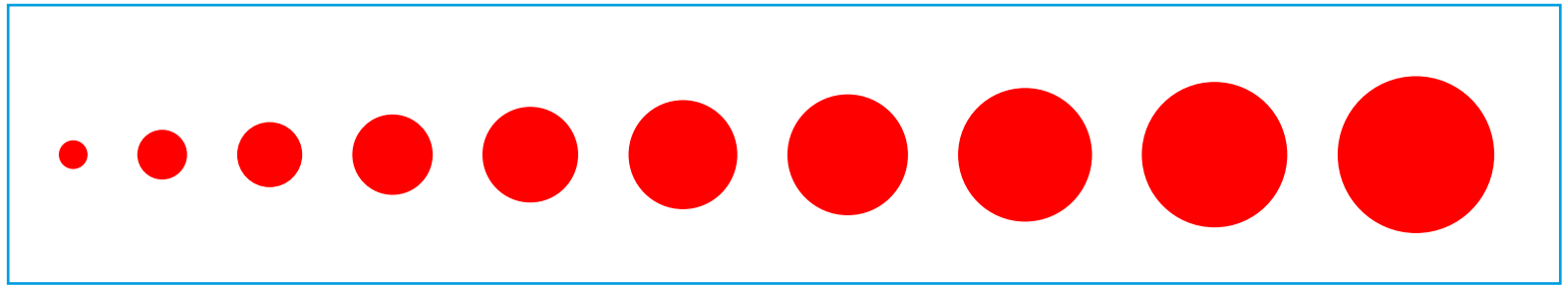
Perception
Nonlinear Scaling

- Application of Stevens's Power Law to the mapping to circle area

- exponent: $a_{area} = 0.7$ ➔ $\Psi_{\text{area}} \propto \mathrm{A}_{\text{circle}}^{0.7} \propto \left(2\pi r_{\text{circle}}^2\right)^{0.7} \propto r_{\text{circle}}^{\frac{7}{5}}$

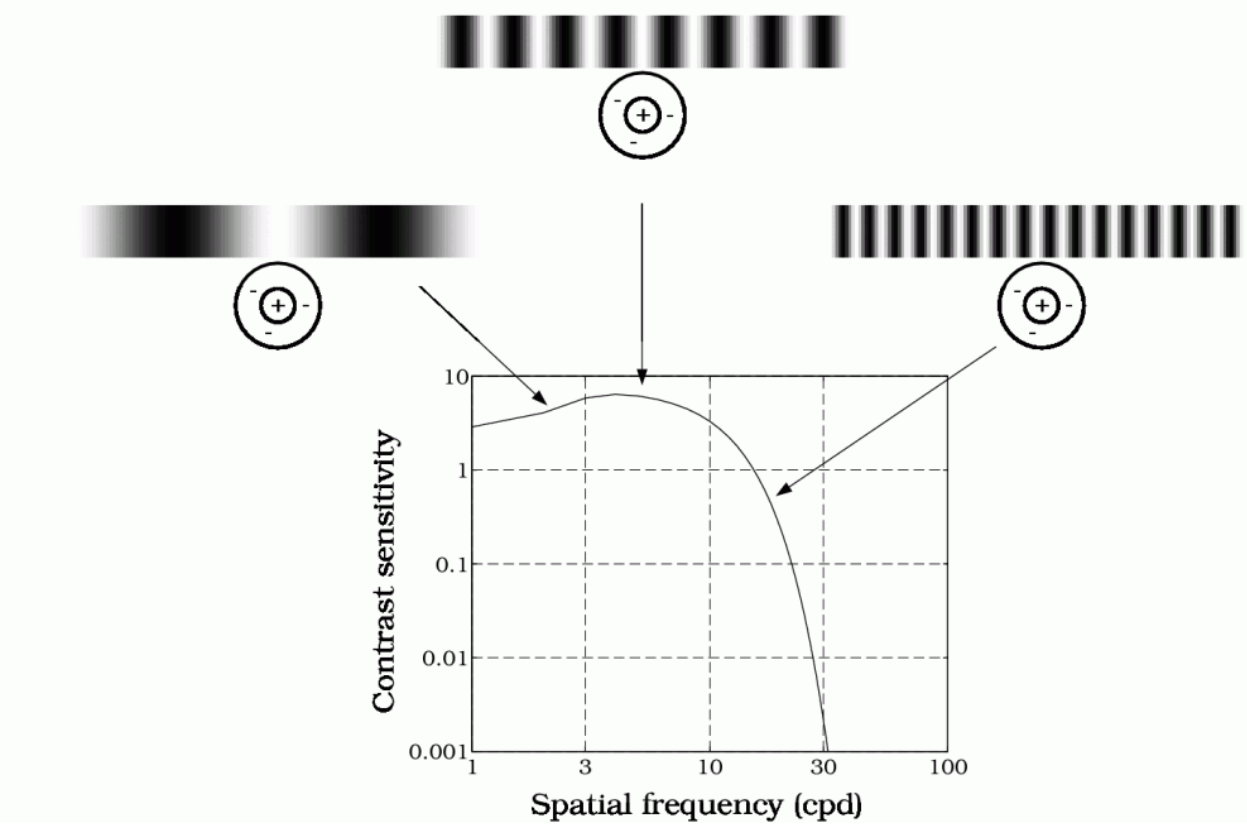
- To scale the perceived area according to a feature f one has to set $f \propto \Psi_{\text{area}} \propto r_{\text{circle}}^{7/5}$.
- Solving for the radius yields

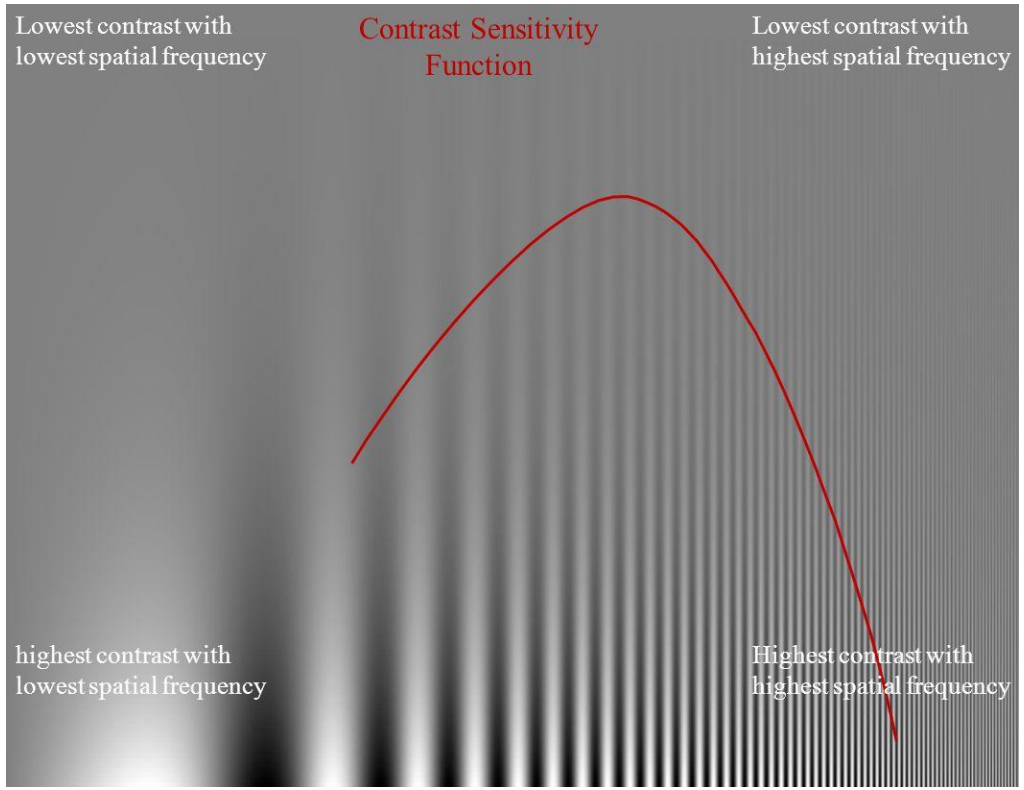
$$r_{\text{circle}} \propto f^{5/7}$$



Example series of circles with the radius increasing according to Stevens's Power Law

Contrast Sensitivity





The graph demonstrates the relationship between contrast sensitivity and spatial frequency. The target-like images are representative of center-surround organization of neurons, with peripheral inhibition at low, intermediate and high angular frequencies. *Used with permission from Brian Wandell, PhD*.

Perception and Cognition

(7) Summary and Outlook

TUD

Summary

You have learned something about that

- visual perception is more than camera work, but a complex cognitive process,
- the visual system is tightly connected to our brain,
- preattentive processing lays the foundation for generating pop-out effects in Vis,
- the Gestalt Principles help us in designing good visualizations,
- color and contrast are not perceived linearly.

Literature

- Colin Ware: *Information Visualization: Perception for Design*, Elsevier/Morgan Kaufmann (3rd ed. 2012)

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- E. Bruce Goldstein: *Sensation and Perception*, Cengage Learning (10th ed. 2017);

  German version: *Wahrnehmungspsychologie: Der Grundkurs*, Springer (9th ed. 2015).

- Preim and Dachselt: *Interaktive Systeme* – Part „Interaktive Informationsvisualisierung“, Springer (2nd ed. 2010)

